

TAFJ-DBTools

R18





Amendment History:

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3	26 October 2012	JN. Charpin	Command execution report
4	15 January 2013	JN. Charpin	R13 review
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10	27 th August 2015	R. Vincent	Promoted column example and further doc
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12	25 th November 2015	H M Hamsa	DBTools security enhancement. Add User inclusion.
13	17 th February	R. Vincent	Add CHANGE-TABLETYPE documentation.



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14	25 th February 2016	R. Vincent	Add CREATE-FILE syntax and update doc.
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16	16 th March 2016	H M Hamsa	DBTools user lockout. tAddUser password complexity
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18	18 th Oct 2016	M.kumar	replace tAddUser with tUserMgnt with –Add/--Reset/--Del options to manage DBTools users.
19	7 th April 2017	R. Vincent	Update table name from LOCK_RECORDS to TAFJ_LOCKS
20	21 st August 2017	R. Vincent	Add MOVE-TABLE documentation
21	24 Nov 2017	M. Kumar	CHANGE-TABLETYPE to include UD type to move into DB files.



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Errata and Comments

If you have any comments regarding this manual or wish to report any errors in the documentation, please document them and send them to the address below:

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DBTools

Introduction

TAFJ-DBTools will be accessed only with authenticated users. Hence username and password has to be passed as parameters to DBTools. The users to access DBTools needs to be created using a file tUserMgmt.bat/tUserMgmt.sh. If there are 3 unsuccessful attempts, user gets blocked. The blocked user cannot be used further until the user is reset with a new password or delete/re-create.

TAFJ-DBTools Console provides access to SQL / JQL / OFS statements, and integrates various TAFJ tools that deal with database within a unique application.

It is also designed to help users to launch various commands in a quick and efficient way by providing aliases and history functionalities. Results are being displayed with a customizable and friendly approach.

The available functionalities are:

- Execute SQL statements
- Execute JQL statements
- Execute OFS statements
- Execute JED application (edit and modify record)
- Extract data from database to file
- Load data from file to database
- Show current locks on database
- Release locks on database
- Display statistics on table



Syntax

Execution – interactive mode

DBTools Console could be executed from TAFJ_HOME/bin.

```
DBTools [-u <username> -p <password> -cf  
         <configuration>[.properties]]
```

The targeted database is the one defined in the default configuration file or in the configuration file specified with the -cf option. The username and password has to be passed along with -u and -p option respectively.

Note:

Copy all jar files from *TAFJ_HOME\dbdrivers\<drivers for different databases>* folder to *TAFJ_HOME\ext folder*.

DBTools Console gives also the ability to launch automatically a default command.

Just add your command to the command line after DBTools Console invocation and it will be executed by default.

```
DBTools [-u <username> -p <password> -cf  
         <configuration>[.properties]] COMMAND
```

i.e.

To launch a SQL command as default command when executing DBTools Console.

```
C:\windows\system32\cmd.exe
D:\Temenos\Development\DEV\T24>DBTools -u tafj -p Tafj@123 SQL SELECT RECID FROM FBK_CURRENCY
```



Execution – script mode

DBTools Console could also execute command in script mode without opening the console. You just have to add '-s' option and the command you want to execute.

```
DBTools [-u <username> -p <password> -s [command] ]
```

i.e.

To script a JQL command

```
C:\windows\system32\cmd.exe
D:\Temenos\Development\DEU\T24>DBTools -u tafj -p Tafj@123 JQL LIST-ITEM FBK.CURRENCY
```

To script a SQL command

```
C:\windows\system32\cmd.exe
D:\Temenos\Development\DEU\T24>DBTools -u tafj -p Tafj@123 -s SQL SELECT RECID FROM FBK_CURRENCY
```

Please note to script a “SELECT * FROM ...” command you will have to escape the * parameter

```
DBTools -u <username> -p <password> -s SQL SELECT \* FROM FBK_CURRENCY
```

Command execution report

When executing a command DBTools could produce different execution reports.

Default mode: execution report buffered in DBTools console

By default without any parameter specified, DBTools console is opened, the command is executed and result is appended to a buffer and displayed in the console at the end of the execution.

Script mode: execution report appended live in OS console

As mentioned above, when providing -s as parameter, “script mode” is enabled, DBTools console is not opened during execution. The execution report is appended live in the OS console.



Log mode: execution report appended in a log file

When providing **-log logFileName** execution report is redirected live to the related log file. This file is generated with .log extension under TAFJ_HOME/log

The “log mode” could be associated to the “script mode” by combining **-s** and **-log logFileName** parameters.

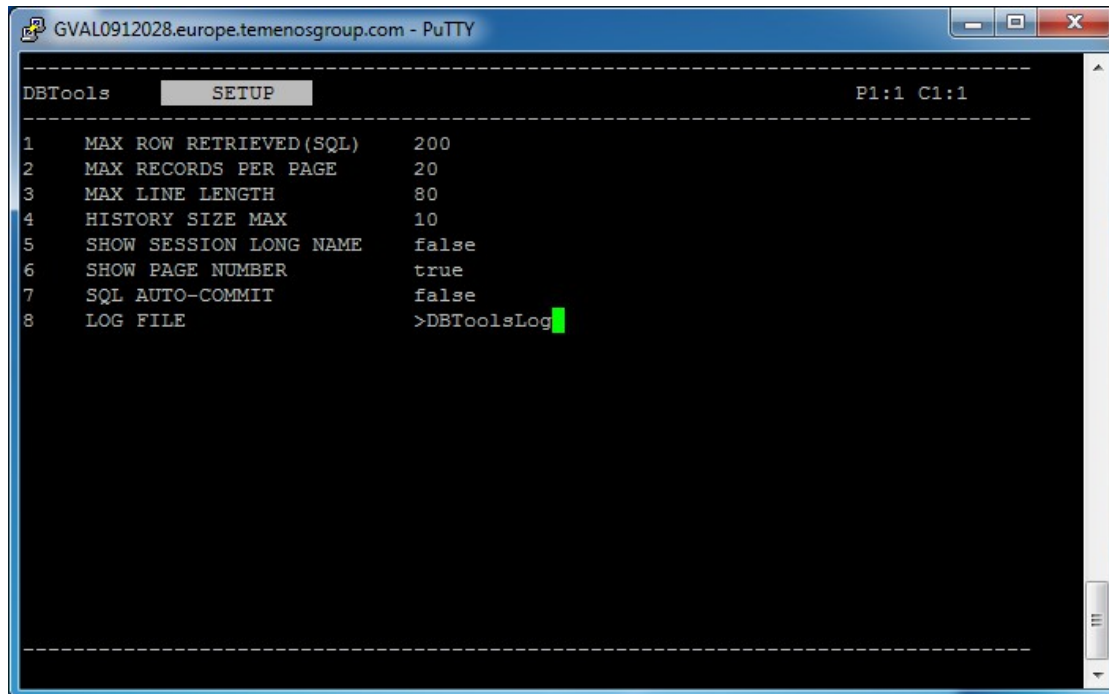
i.e.

```
DBTools -u <username> -p <password> -s -log currencyId SQL
SELECT RECID FROM FBNK_CURRENCY
```

In that case all FBNK_CURRENCY IDs will be outputted to the OS console and to currencyId.log under TAFJ_HOME/log.

The “log mode” could be associated to the “console mode” by providing a log file name in the DBTools console, from setup menu.

The setup menu is available by typing **setup** on the command line. Please refer to the setup section for more information about it.



Once the log file has been setup it's necessary to turn on the **SPOOL** feature to tell the console to output the result to the log file.



It could be done by typing **SPOOL** on the command line. You will see that SPOOL is ON in the header of the console. By retyping **SPOOL** you will turn it off and disable the logging functionality.

This way you could choose which command execution report you want to append to log file.

```
GVAL0912028.europe.temenosgroup.com - PuTTY
DBTools  SETUP  SPOOL ON  P1:1 C1:1
-----
1  MAX ROW RETRIEVED (SQL)  200
2  MAX RECORDS PER PAGE   20
3  MAX LINE LENGTH        80
4  HISTORY SIZE MAX       10
5  SHOW SESSION LONG NAME false
6  SHOW PAGE NUMBER       true
7  SQL AUTO-COMMIT        false
8  LOG FILE               DBToolsLog
-----
tafj@localhost:DEV>SPOOL
```

Silent mode: no execution report

When providing parameter **-silent** in “script mode”, command is executed silently, no output is generated.

i.e.

```
DBTools -u <username> -p <password> -s -silent COMMAND
```



Presentation

DBTools console consists of three panels:

- Header panel that contains information about the current mode or running command
- Main panel that contains the result of the last executed command
- Footer panel that contains the command line

The screenshot shows a terminal window titled "GVAL0912028.europe.tamagroup.com - PuTTY". The console is divided into three horizontal panels, each highlighted with a red border:

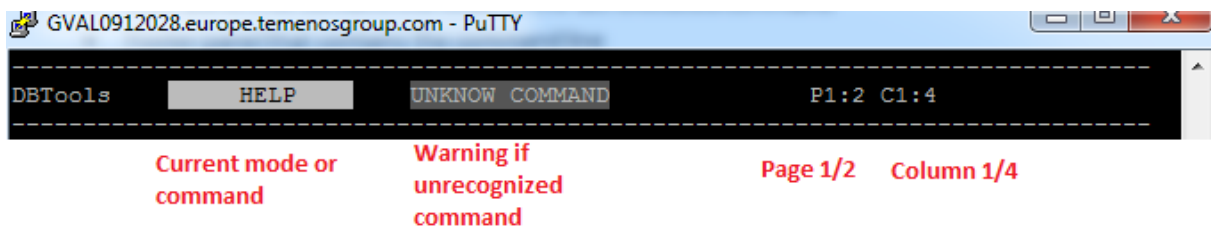
- Header panel:** Contains the text "DBTools", a "HELP" button, the word "Header", and the coordinates "P1:2 C1:4".
- Main panel:** Contains a list of supported commands and their usage instructions, such as "SQL commands are supported | type SQL to enter SQL mode", "JQL commands are supported | type JQL to enter JQL mode", "JED commands are supported | type JED <<FILE NAME>> <<RECORD KEY>> to open JED C", "Data management is supported | type EXTRACT-TABLE <<FILE NAME>> <<path>> file mode", "Lock management is supported | type LOCKS to see records locked", "Statistics are supported | type STAT <<FILE NAME>>", and "ALIASES commands | Type alias <<alias name>> <<command>> to create an".
- Footer panel:** Contains the command prompt "tafj@Oracle:localhost>" and the word "Footer".



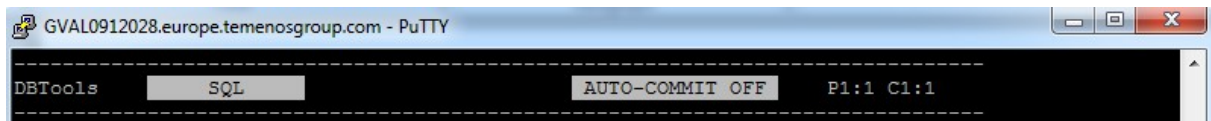
Header

The header presents the following information:

- The current mode or running command
- A warning in case of unrecognized command
- The current page / column number and the total number of page / column if the result of the command is being displayed on multiple page and/or columns.



In SQL Mode, you will also see the auto-commit status. Please refer to SQL and setup sections to get more information on that point.



Main panel

The main panel contains the command result to be displayed or edited. It is displayed vertically and split into lines and eventually columns.

If the data is larger than the console maximal length (80 characters by default) it will be displayed on multiple columns.

If the result has more lines than the console could support (20 rows by default) it will be displayed on multiple pages.

i.e. SQL Select result that is being displayed on 2 pages and 12 columns.



```
GVAL0912028.europe.temenosgroup.com - PuTTY

DBTools  SQL  P1:2 C1:12

-----
RECID|THE_RECORD|CURRENCY_CODE|RANK|NUMERIC_CCY_CODE|CCY_NAME|NO_OF_DE
-----
AUD  |          |AUD          |    |30          |AUSTRALIAN DOLLAR|2
CAD  |          |CAD          |    |50          |CANADIAN DOLLAR  |2
CHF  |          |CHF          |6   |60          |SWISS FRANCS     |2
CNY  |          |CNY          |    |388         |Chinese Yuan     |2
EUR  |          |EUR          |5   |100         |Euro             |2
GBP  |          |GBP          |1   |130         |POUND STERLINGS  |2
GOL  |          |GOL          |15  |6           |BULLION1         |2
HKD  |          |HKD          |    |150         |HONK KONG DOLLARS|2
INR  |          |INR          |    |310         |INDIAN RUPEE     |2
JPY  |          |JPY          |4   |180         |JAPANESE YEN     |0
KHR  |          |KHR          |    |33          |riel (KHR)       |2
KRW  |          |KRW          |    |389         |KOREAN WON       |0
LBP  |          |LBP          |    |270         |Lebanese Pounds  |2
NZD  |          |NZD          |    |220         |New Zealand      |2
PLN  |          |PLN          |    |280         |Polish Zloty     |2
SEK  |          |SEK          |    |240         |Swedish Krone    |2
XEU  |          |XEU          |    |954         |Euro             |2
ZAR  |          |ZAR          |    |290         |South African Rand|2
SGD  |          |SGD          |    |250         |SINGAPORE DOLLAR |2
-----

tafj@Oracle:localhost>
```

For command results that can be edited, i.e. console setup parameters, a field number is displayed on a margin on the left part of the panel; the associated value is displayed next to the margin on the middle of the panel.

It's then possible to access the related data by typing its associated field number on the command line. Please refer to Setup functionality section for more information.

i.e. Setup command result that can be edited, by typing '2' you will be able to change actual value '20' to another value.

```
GVAL0912028.europe.temenosgroup.com - PuTTY

DBTools  SETUP  P1:1 C1:1

-----
1  MAX ROW RETRIEVED(SQL)  200
2  MAX RECORDS PER PAGE   20
3  MAX LINE LENGTH        80
4  HISTORY SIZE MAX       10
5  SHOW SESSION LONG NAME false
6  SHOW PAGE NUMBER       true
7  SQL AUTO-COMMIT        false
8  LOG FILE                DBToolsLog
-----

tafj@127.0.0.1:TAFJDB>
```



Footer

The footer panel contains the command line that shows by default the database connection parameters: user name, database provider and hostname.

```
-----  
tafj@Oracle:localhost> █
```

Depending on console setting it could also show the hostname ip address and the database name.

```
-----  
tafj@Oracle:192.168.1.11:1521:TBR11GA> █
```

Database Functionalities

SQL

To execute SQL statements, first you have to enter **SQL mode** by typing command “**SQL**” on the command line or **by prefixing your SQL command with SQL keyword**.

```
-----  
tafj@Oracle:localhost> SQL █
```

Then you will see that SQL mode is activated in the header panel and the auto-commit status.

```
GVAL0912028.europe.temenosgroup.com - PuTTY  
-----  
DBTools  SQL  AUTO-COMMIT OFF  P1:1 C1:1  
-----
```

The console is now ready to execute SQL statements.

To be recognized the statement must start with one of the following keywords in uppercase:

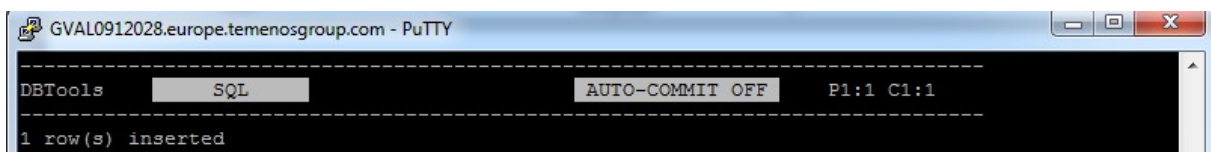
- SELECT
- INSERT
- UPDATE
- DELETE



- CREATE
- ALTER
- DROP
- COMMIT
- ROLLBACK
- DESCRIBE

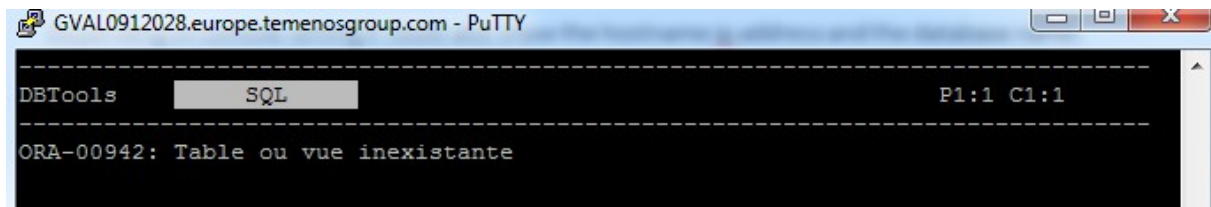
By default the database connection is set to auto-commit=false, that means if you don't apply the COMMIT statement after your data modification, your change will be roll backed when exiting the console.

Once your statement has been executed, the result will be displayed on the console if that statement has got a result set, otherwise you will get information about the number of rows affected.



```
-----  
DBTools      SQL      AUTO-COMMIT OFF      P1:1 C1:1  
-----  
1 row(s) inserted
```

In case of SQL Exception the related message will be displayed on the console.

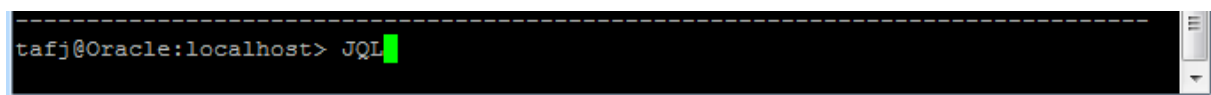


```
-----  
DBTools      SQL      P1:1 C1:1  
-----  
ORA-00942: Table ou vue inexistante
```

To avoid an excessive memory usage when retrieving important set of data, by default only the first 200 rows will be displayed. You can change this parameter to the value you want, please refer to Setup section to do so.

JQL

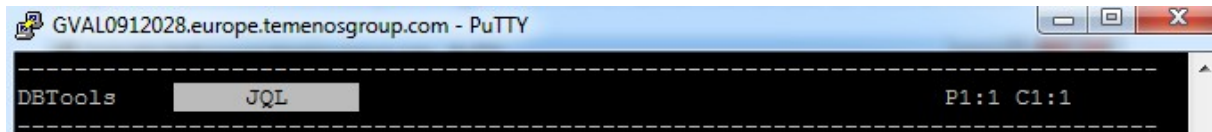
To execute JQL statements, first you have to **enter JQL mode** by typing command “JQL” on the command line or by **prefixing your JQL command with JQL keyword**.



```
-----  
tafj@Oracle:localhost> JQL
```



Then you will see that JQL mode is activated in the header panel.



The console is now ready to execute JQL statements.

To be recognized the statement must start with one of the following keywords in uppercase:

- COPY <<Source Table Name>> <<Destination Table Name>> <<ReclId optional>>
- COUNT <<VOC File name>>
- GETDBNAME <<VOC File name>>
- LIST-ITEM <<VOC File name>> <<WITH clause>>
- DESCRIBE <<View or Table Name Pattern>> <<Column Name Pattern (optional)>>...ie. TAFJV_FBNK_CURR% %CURR%
- LOCATE-TABLE <<Table Name>> (zOS only)
- CLEAR-FILE <<VOC File name>>
- CREATE-FILE <<VOC File name>>
 - o Several different syntaxes can be used.
 - o To create a table with a particular table type (XML,CLOB,BLOB,MULTI-COLUMN,VARCHAR), use the syntax:
 - CREATE - FILE F.TESTR0FILE TYPE=<<TYPE>>
 - o For read-only and \$DIM files:
 - Two commands are needed to create new T24 archive tables.
 - CREATE - FILE F.TESTR0FILE TYPE=XML ASSOCIATE="YES"
 - The command above creates the table in the transactional database and sets it up for a "read-only" pair.
 - CREATE - FILE F.TESTR0FILE\$R0 TYPE=XML
DATABASE="TESTRODB" SCHEMA="TAFJR0" READONLY="YES"
 - o To specify MULTI-COLUMN file creation, an existing dictionary is needed, ie.
CREATE-FILE FBNK.CURRENCY.2 TYPE=MULTI-COLUMN
DICT=D_F_CURRENCY



- DELETE-FILE <<VOC File name>>
- SHOWDICTS <<TYPE= (I|D|PH)>> <<ATTRIBUTE= (ALL| Attribute Number to Search)>> <<(STARTSWITH|ENDSWITH|CONTAINS)="Search String">> <<FILE= FileName to log.>>

SHOWDICTS allows you to search all dictionary items. Logs show up in \$TAFJ_HOME/log. If the FILE parameter is not specified, the default ShowDicts.txt is used.

This example below shows any dict item I type with ATTRIBUTE 2 containing the search "ENQ.TRANS":

```
DBTools>SHOWDICTS TYPE=I ATTRIBUTE=2 CONTAINS="ENQ.TRANS"
FILE=enqtrans.txt
```

The output file \$TAFJ_HOME/log/enqtrans.txt would look like:

```
FMF1.MD.DEAL FMF1_MD_DEAL D_F_MD_DEAL CUST.SECTOR I?CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "SECTOR")??CUST.SECTOR?4R?S????????CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "SECTOR")

FMF1.MD.DEAL FMF1_MD_DEAL D_F_MD_DEAL INDUSTRY I?CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "INDUSTRY")??INDUSTRY?4R?S????????CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "INDUSTRY")

FMF1.MD.DEAL FMF1_MD_DEAL D_F_MD_DEAL RESIDENCE I?CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "RESIDENCE")??RESIDENCE?2L?S????????CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "RESIDENCE")

FMF1.MD.DEAL FMF1_MD_DEAL D_F_MD_DEAL SECTOR I?CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "SECTOR")??SECTOR?4R?S????????CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "SECTOR")

FMF1.MD.DEAL FMF1_MD_DEAL D_F_MD_DEAL TARGET I?CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "TARGET")??TARGET?4R?S????????CUSTOMER;
SUBR("ENQ.TRANS","CUSTOMER", @1, "TARGET")
```

This format is easily imported into Excel with tab delimiters.

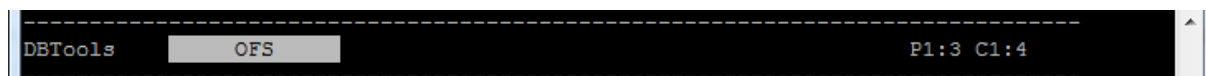
OFS

To execute OFS statements, first you have to **enter OFS mode** by typing command "**OFS**" on the command line or by **prefixing your OFS command with OFS keyword**.

```
tafj@localhost:TAFJDB>OFS
```

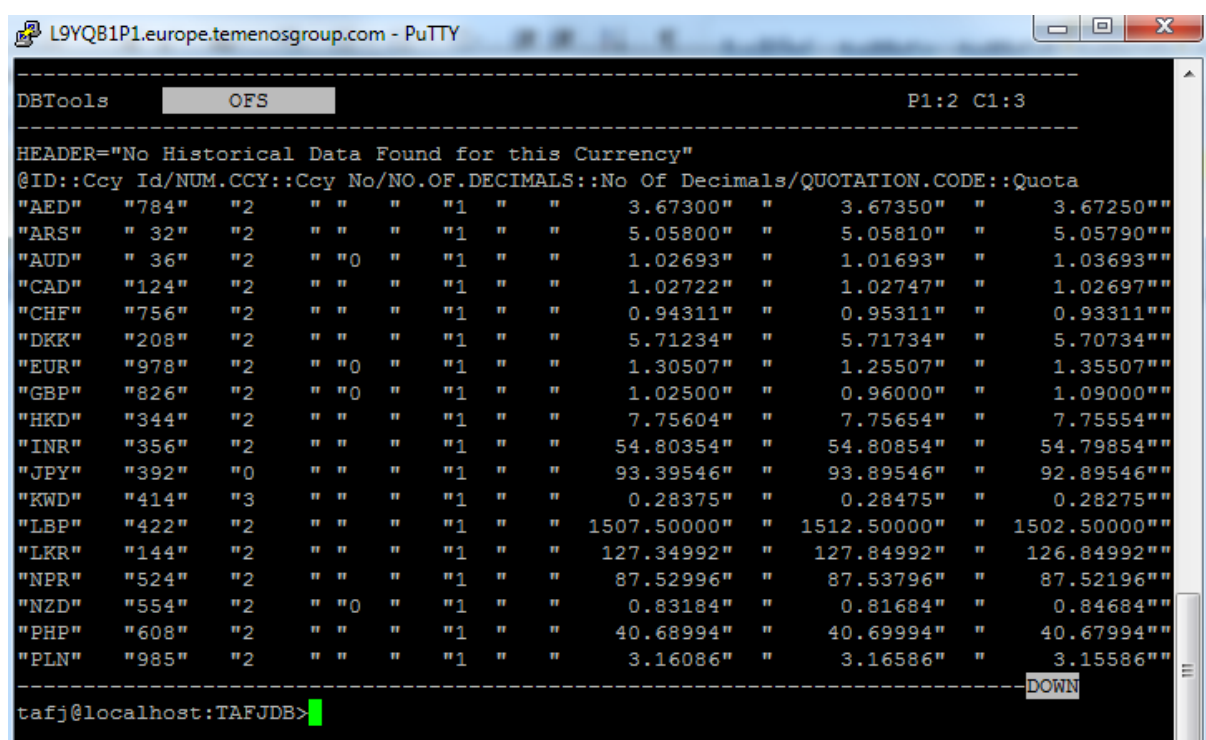
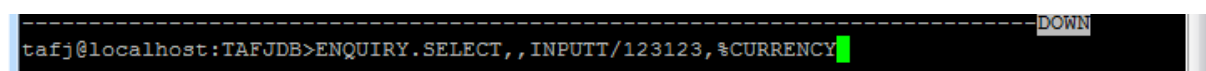


Then you will see that OFS mode is activated in the header panel.



The console is now ready to execute OFS statements, please note that your TAFJ properties file must point to a valid T24 precompile directory or jars.

The default OFS source is GCS.





MOVE- TABLE

MOVE-TABLE statement will allow you to move tables from one schema to another schema within the same database. Both the table and view will be moved. Both schemas must already exist. If the table is part of a JOIN view, then that view becomes invalidated. This command will not work if the target view is part of a JOIN view, so please check that it isn't.

Syntax

The MOVE-TABLE syntax is the following:

```
MOVE-TABLE TABLE=<<J4FileName>> FROM=<<SCHEMA>> TO=<<SCHEMA>>
```

CHANGE- TABLETYPE

CHANGE-TABLETYPE statements will allow you to change a table type from one type of data storage to another (example, from XML to CLOB, or vice-versa) while moving the existing data to the new format. A backup file is created with the original data named with the suffix .BACKUP of the target file. This backup file is deleted by default. **Dictionary tables are not supported. In addition, types currently supported are XML, CLOB, BLOB, and VARCHAR,DIRECTORY at the moment.**

Syntax

The CHANGE-TABLETYPE syntax is the following:

```
CHANGE-TABLETYPE [options] TABLE=<<J4FileName|ALL(only valid when  
FROM is type=DIRECTORY, to push all UD's into DB)>> FROM=<<CLOB|  
BLOB|XML|VARCHAR|MULTI-COLUMN|DIRECTORY >> TO=<<CLOB|BLOB|XML|  
VARCHAR|MULTI-COLUMN>>
```

The only valid option is:

-nd : don't delete backup of old file.

Example

```
CHANGE - TABLETYPE TABLE=FBNK.CURRENCY FROM=XML  
TO=CLOB
```

The resulting output looks like this:



```
-----
DBTools      CHANGE-TABLETYPE                               P1:1 C1:1
-----
Starting-->
Dropping file FBNK.CURRENCY.BACKUP
Creating file FBNK.CURRENCY.BACKUP
Copying to backup file
Dropping file: FBNK.CURRENCY
Creating file: FBNK.CURRENCY with type: CLOB
Copying data from backup to file: FBNK.CURRENCY
Dropping file FBNK.CURRENCY.BACKUP
Completed.

-----
tafj@localhost:TB201507>
```

DBCHECK

You can execute the following commands to do checks on your database by typing DBCHECK to put DBTools in DBCHECK mode, then USAGE to arrive at the below screen:



```
tafj@benchbox: ~/TAFJ/DEV_201304/bin
DBTools  DBCHECK  P1:1 C1:4
-----
DBCheck Commands      |The following commands are supported....
<command>|<command2>|<command3>|Allows the user to specify more than one command
CHECK NUMERICS        |Iterate the dicts searching for numerics defined
CHECK LENGTH          |Check length for Dtypes against maximum length i
CHECK MISSING TABLE  |CHECKS TAFJ_VOC to ensure each table referenced

tafj@localhost:R12GAMB1MILL>
```

All commands log to a particular file that is shown to the user once the command runs.

CHECK NUMERICS iterates the dictionary items where dictionary items are defined as numeric. It then tests those fields to make sure the data is indeed numeric. As well, it tests those VOC items that are defined rightJustified to see if the data is numeric.

CHECK LENGTH iterates the dictionary items and gathers defined lengths and checks each against the maximum length stored in the column. It reports anything that is incorrectly defined. THIS COMMAND TAKES A LONG TIME TO RUN (More than 3 hours depending on the size of the DB).

CHECK MISSING TABLE iterates VOC items and tests that the tables exist in the database.



JED

You can call JED console to display and edit database records by using command JED <<FILE NAME>> <<RECORD KEY>>

```
tafj@Oracle:localhost> JED F.TSA.SERVICE COB
```

```
-----  
File F.TSA.SERVICE, Record COB  UNLOCKED  
-----  
1      FOR COB  
2  
3      ONE  
4  
5  
6      STOP  
7  
8  
9  
10  
11  
12  
13  
14  
15.1   20091222  
15.2   20091222  
15.3   20091222  
16.1   20091226  
16.2   21/06/2010 09:22:43  
16.3   01/02/2010 21:06:25  
-----  
Command({l},i{l},d{l},b,f,c,r,s,x,help) :
```

You can also call JED -s <<FILE NAME>> <<RECORD KEY>> <<FIELD NUMBER>>=<<VALUE>>; to directly update a record without opening JED Console.

```
tafj@localhost:TESTDB>JED -s F.TSA.SERVICE COB 6=START;
```

For multiple attributes, separate them with “;”.

```
tafj@localhost:TESTDB>JED -s F.TSA.SERVICE COB 3=TWO;6=START;
```

Depending on the operating system you could have to escape the part of command after -s option with quotes: JED -s 'F.TSA.SERVICE COB 6=START;'

Please refer to the JED Console section at the end of this document to get more information about JED Console usage.



Extract table

This command allows extracting the records from a table and writing them on the local file system. Depending on the option you choose, the result will be:

-f : file mode (default). A set of distinct file, one per record, the name of the file has the format <filename>-<recordId>

-d : directory mode. A directory having the name of the table and inside this directory each record will be stored in a distinct file, the name of this last file will be the actual record Id.

If any file or directory already exists, they will be simply overridden.

To extract the CURRENCY table as a set of records, just type

```
tafj@Oracle:localhost> EXTRACT-TABLE FBNK.CURRENCY /home/user/currency
```

This will result in a directory structure like this one:

/home/user/currency

FBNK.CURRENCY -GBP

FBNK.CURRENCY -CHF

FBNK.CURRENCY -USD

...

Where FBNK.CURRENCY -GBP, FBNK.CURRENCY -CHF ... are files containing the record.

All extracted files extracted and eventually raised exceptions will be shown in the console as result.



```
GVAL0912028.europe.temenosgroup.com - PuTTY
-----
DBTools  EXTRACT TABLE  P1:1 C1:1
-----
Creating c:\temp\currency\FBNK.CURRENCY
Processing : AUD
Processing : CAD
Processing : CHF
Processing : CNY
Processing : EUR
Processing : GBP
Processing : GOL
Processing : HKD
Processing : INR
Processing : JPY
Processing : KHR
Processing : KRW
Processing : LBP
Processing : NZD
Processing : PLN
Processing : SEK
Processing : SGD
Processing : USD
Processing : XEU
Processing : ZAR

Nb Records : 20
Nb Errors : 0
Done in : 78 ms.

-----
tafj@Oracle:localhost> 
```

To extract the CURRENCY table as a set of records in their own directory, just type

```
tafj@Oracle:localhost> EXTRACT-TABLE -d FBNK.CURRENCY /home/user
```

This will result in a directory structure like this one:

/home/user/

 FBNK.CURRENCY/

 GBP

 CHF

 USD

 ...

Where GBP, CHF ... are files containing the record.



All extracted files extracted and eventually raised exceptions will be shown in the console as result.

Known issue: as it is possible that the record id of the record to be extracted contain the character “/”, extraction will fail as we can’t have “/” in a file name.

Load data

This command loads data from your file system into your database. You can provide as parameter a file to load a single record, or a directory to load a set of records i.e.:

-f : file mode (default) . The file name has the format <filename>-<recordId>
If the file passed is a directory, all files having the previous format will be processed

-d : directory mode. The parameter must be a directory.
This directory will be recursively processed until a file is found.

The parent directory of this file represents the FileName, file name itself is the recordId

To load a single record from a file on your file system

```
tafj@Oracle:localhost> LOAD-DATA /home/user/currency/FBNK.CURRENCY-USD
```

You will get as result displayed in the console the table impacted and the record id inserted.

```
GVAL0912028.europe.temenosgroup.com - PuTTY
-----
DBTools      JQL                                     P1:1 C1:1
-----
Table : FBNK.CURRENCY / Id :USD
```

To load a set of records from a directory on your file system

```
tafj@Oracle:localhost> LOAD-DATA -d /home/user/FBNK.CURRENCY
```



```
GVAL0912028.europe.temenosgroup.com - PuTTY
DBTools  LOAD DATA  P1:1 C1:1
-----
Table : FBNK.CURRENCY / Id :AUD
Table : FBNK.CURRENCY / Id :CAD
Table : FBNK.CURRENCY / Id :CHF
Table : FBNK.CURRENCY / Id :CNY
Table : FBNK.CURRENCY / Id :EUR
Table : FBNK.CURRENCY / Id :GBP
Table : FBNK.CURRENCY / Id :GOL
Table : FBNK.CURRENCY / Id :HKD
Table : FBNK.CURRENCY / Id :INR
Table : FBNK.CURRENCY / Id :JPY
Table : FBNK.CURRENCY / Id :KHR
Table : FBNK.CURRENCY / Id :KRW
Table : FBNK.CURRENCY / Id :LBP
Table : FBNK.CURRENCY / Id :NZD
Table : FBNK.CURRENCY / Id :PLN
Table : FBNK.CURRENCY / Id :SEK
Table : FBNK.CURRENCY / Id :SGD
Table : FBNK.CURRENCY / Id :USD
Table : FBNK.CURRENCY / Id :XEU
Table : FBNK.CURRENCY / Id :ZAR
```

Show Locks

By launching LOCKS command you will be able to see all records locked on the database. It monitors locks managed by tLockManager, the TAFJ_LOCKS table (JDBC Locking), or ORCL, or MSQL, or DATABASE internal locking on the database. Below is a screenshot from tLockManager.

```
tafj@Oracle:10.41.2.22:1521:TBR11GA> LOCKS
```

The result displayed is the following.

Note that MSQL LockManager will crash Eclipse with sqljdbc_4.1 drivers and will not produce the right results (DBTools LOCKS command). It should also be noted that sp_lock command truncates data. Version sqljdbc_1.2 of drivers have been tested and work.



```
Administrator: CMD - DBTools -cf DEVPROC -u mj -p 123456

DBTools      SHOW LOCKS      P1:1 C1:1
-----
File name    |Record  |SessionId
FBNK_SECTOR#NAU|    1001 | 97775675

t24@localhost:DEV>
```

This is different than what you might see with DATABASE locking. When issuing the LOCKS command, non-existent rows will appear as NonExist*<<TAFJ Session Unique ID>> for the Session Id column, whereas one that was taken via SELECT FOR UPDATE, will show the Oracle Session ID for the Session Id column. One cannot release locks taken with SELECT FOR UPDATE. The Oracle session must be either committed, rolled back, or killed. As well, tables with more than 25,000 rows will show up as F_PGM_DATA_CONTROL *UNKNOWN, if F_PGM_DATA_CONTROL had more than 25,000 rows. This is because Oracle has no way to report row level locking except by iterating all of the rows and attempting to lock them. This is very costly in terms of performance, and hence, why TAFJ caps the number of rows.

```
-----
DBTools      SHOW LOCKS      P1:1 C1:1
-----

LOCK ID          |Session Id
FBNK_CURRENCY*ZZZ      |NonExist*1193893622
FBNK_CATEG_ENTRY*169376209561194.050001|128
```

This is also different than what you might see with ORCL, or MSSQL locking. Below is a screenshot of ORCL locking. The 53 below denotes the *database* session id that *called* the



lock. 205121125 represents the TAFJ unique session for a thread. So now TAFJ can match a java thread with the database session that locked it.

```
-----  
-DOWN  
tafjuser@10.41.5.54:TESTDB>LOCKS  
-----  
-----  
DBTools          SHOW LOCKS                                P1:1  
C1:1  
-----  
-----  
LOCK ID          |Session Id  
FBNK_CURRENCY*USD|2051521125*53  
-----
```

Keep in mind that there is one connection for a thread actually asking for a lock, and another connection that actually takes the lock (this is due to auto-committing of the lock connection). What is reported is the database session id for the lock connection which will also appear in the locking.log (for JDBC Locking it will only appear in the log).

```
[DEBUG] 2015-04-02 15:18:40,656 [main] LOCKING - LOCKING  
FBNK_CURRENCY*USD / 2051521125*53...
```

```
[DEBUG] 2015-04-02 15:18:40,661 [main] LOCKING - LOCK FBNK_CURRENCY*USD /  
2051521125*53 -> OK
```

Release Locks

The release locks command is specially designed for tLockManager. It is used to release the Locked record from tLockManager.

You can check which records are currently locked by launching command LOCKS and then release the record you want by typing command RELEASE <<FILE NAME>> <<RECORD ID>>



```
GA. Administrator: CMD - DBTools -cf DEVPROC -u mj -p 123456

DBTools      SHOW LOCKS      P1:1 C1:1
-----
File name    |Record| |SessionId|
FBNK_SECTOR#NAU|    1001| | 97775675|

t24@localhost:DEV>RELEASE FBNK_SECTOR#NAU_1001
```

Result being displayed after releasing the lock.

```
GVAL0912028.europe.temenosgroup.com - PuTTY

DBTools      RELEASE LOCKS      P1:1 C1:1
-----
1 lock(s) released

tafj@Oracle:10.41.2.22:1521:TBR11GA> LOCKS
```



```
GVAL0912028.europe.temenosgroup.com - PuTTY
-----
DBTools  SHOW LOCKS  P1:1 C1:1
-----
No Current Lock

-----
tafj@Oracle:10.41.2.22:1521:TBR11GA> █
```

Note: With ORCL and MSQL locking, it is not possible to release locks because DBTools does not own the database session where the lock was taken. As well, with DATABASE locking, it is not possible to release locks that have been taken with SELECT FOR UPDATE (see above SHOW LOCKS section for more details) for the same reason.

Database statistics

This command allows to easily getting information about your JBase file, JBase file path, JBase equivalent database filename, created date and record count.

To get information about the file you have configured type STAT <<FILE.NAME>>

```
tafj@Oracle:localhost> STAT FBNK.CURRENCY █
```

The resulting output looks like this:

```
GVAL0912028.europe.temenosgroup.com - PuTTY
-----
DBTools  DB STAT  P1:1 C1:1
-----
JBase File  |FBNK.CURRENCY
File Path   |../bnk.data/st/FBNK_CURRENCY
Oracle File  |FBNK_CURRENCY
Created Date |20-juin-2011
Record Count |20
-----
```



Console Functionalities

HELP

When opening DBTools Console the “HELP” command will launched by default and the available commands will be displayed on the main panel.

```
GVAL0912028.europe.temenosgroup.com - PuTTY
DBTools  HELP  P1:2 C1:4
-----
SQL  commands are supported |type SQL to enter SQL mode
JQL  commands are supported |type JQL to enter JQL mode
JED  commands are supported |type JED <<FILE NAME>> <<RECORD KEY>> to open JED C
      |type JED <<FILE NAME>> <<RECORD KEY>> <<FIELD NUMBE
Data management is supported|type EXTRACT-TABLE <<FILE NAME>> <<path>> file mode
      |type EXTRACT-TABLE -d <<FILE NAME>> <<path>> direct
      |type LOAD-DATA <<file>> file mode (default), the fi
      |type LOAD-DATA -d <<directory>> directory mode. T
Lock management is supported|type LOCKS to see records locked
      |type RELEASE <<FILE NAME>> <<RECORD>>
Statistics are supported    |type STAT <<FILE NAME>>
HELP command                |Type help to display the available commands and the
ALIASES commands            |Type alias <<alias name>> <<command>> to create an
      |Type unalias <<alias name>> to delete an existing a
-----
tafj@Oracle:localhost> █
```



```
GVAL0912028.europe.temenosgroup.com - PuTTY
-----
DBTools      HELP      P2:2 C1:4
-----
|Type list-aliases to display all defined aliases
|
HISTORY commands |Type history to get last used commands
|Type h<<command number>> to recall command stored a
|Type clear-history to delete all stored commands
|
SETUP commands   |Type setup to change console and session parameters
|
NAVIGATION commands |Type f to go forward to previous page
|Type b to go backward to previous page
|Type sr to scroll right to next column
|Type sl to scroll left to previous column
|Type cm to disable\enable column mode
|Type hc to disable\enable first fixed result column
|Type hl to disable\enable first fix result line
|Type x to exit
-----
tafj@Oracle:localhost> █
```

You can access this screen whenever you want by typing command “help” on the command line.

```
tafj@Oracle:localhost> help█
```

History

A history of last commands launched (10 last commands by default) is maintained to give the ability to reuse quickly last used command.

Simply enter “history” on the command line.

```
tafj@Oracle:localhost> history█
```

You will get the resulting output that shows the list of last used command.



```
GVAL0912028.europe.temenosgroup.com - PuTTY

DBTools  HISTORY  P1:1 C1:1

-----
History|Last command used
h1      |SQL SELECT * FROM TAFJV_FBNK_CURRENCY
h2      |EXTRACT-DATA FBNK.CURRENCY c:\temp\currency
h3      |SQL SELECT * FROM FBNK_CURRENCY
h4      |LOAD-DATA c:\temp\currency\FBNK.CURRENCY-USD
h5      |JQL LIST FBNK.CURRENCY
h6      |JQL COUNT FBNK.CURRENCY
h7      |STAT FBNK.CURRENCY
h8      |RELEASE F_TSA_SERVICE COB
h9      |LOCKS
h10     |LOAD-DATA -d c:\temp\currency

-----

tafj@Oracle:localhost> h1
```

Whether you are under history mode or not you can recall the command you want by typing its history index on the command line. i.e. you want to recall last SQL command just type h1 and you will get that command executed.



```
GVAL0912028.europe.temenosgroup.com - PuTTY

DBTools  SQL  P1:2 C1:12

-----
RECID|THE_RECORD|CURRENCY_CODE|RANK|NUMERIC_CCY_CODE|CCY_NAME        |NO_OF_DE
GBP  |          |GBP           |1   |130          |POUND STERLINGS    |2
GOL  |          |GOL           |15  |6            |BULLION1           |2
HKD  |          |HKD           |    |150          |HONK KONG DOLLARS  |2
INR  |          |INR           |    |310          |INDIAN RUPEE       |2
JPY  |          |JPY           |4   |180          |JAPANESE YEN       |10
KHR  |          |KHR           |    |33           |riel (KHR)         |2
KRW  |          |KRW           |    |389          |KOREAN WON         |10
LBP  |          |LBP           |    |270          |Lebanese Pounds    |2
NZD  |          |NZD           |    |220          |New Zealand        |2
PLN  |          |PLN           |    |280          |Polish Zloty       |2
SEK  |          |SEK           |    |240          |Swedish Krone      |2
XEU  |          |XEU           |    |954          |Euro               |2
SGD  |          |SGD           |    |250          |SINGAPORE DOLLAR   |2
USD  |          |USD           |10  |260          |US Dollar1         |2
ZAR  |          |ZAR           |    |290          |South African Rand|2
AUD  |          |AUD           |    |30           |AUSTRALIAN DOLLAR  |2
CAD  |          |CAD           |    |50           |CANADIAN DOLLAR    |2
CHF  |          |CHF           |6   |60           |SWISS FRANCS       |2
CNY  |          |CNY           |    |388          |Chinese Yuan       |2
-----

tafj@Oracle:localhost>
```

Please note that history is saved at the end of your session so you have the ability to load it the next time you will use DBTools Console. Data are stored in a properties file “DBTools” under TAFJ_HOME/conf.

If you don’t want to keep that history for your next session, or to avoid other user to get it, as the DBTools properties file is shared between all users, you can clear it by typing command “clear-history”.

```
tafj@Oracle:localhost> clear-history
-----
```

As mentioned before DBTools Console doesn’t rely on a user context, in consequence if many users use it at the same time the last one who quit his session will override the properties file with his settings.

Aliases

Create an alias

DBTools Console also provides an “alias” functionality to reuse complex command efficiently.

For example you often use the same JED statement to setup F.TSA.SERVICE, record COB, status to “START” and number of AGENT to “TEN” : JED F.TSA.SERVICE COB 3=TEN;6=START;



You can create an alias to create a shortcut to this command by typing :

alias <<ALIAS NAME>><<BLANK>><<COMMAND>>

```
tafj@Oracle:localhost> alias STARTCOB JED F.TSA.SERVICE COB 3=TEN;6=START;
```

Display aliases

To display all stored aliases you can type command "aliases"

```
GVAL0912028.europe.temenosgroup.com - PuTTY

DBTools  ALIASES  P1:1 C1:1
-----
Alias    |Command
STARTCOB|JED F.TSA.SERVICE COB 3=TEN;6=START;

tafj@Oracle:localhost> aliases
```

Execute an alias

To execute this command just type the alias name on the command line.

```
tafj@Oracle:localhost> STARTCOB
```

Please note that if you create an alias for a SQL or JQL statement you have to switch to the appropriate mode before calling the alias or the command won't be recognized.

i.e.

```
tafj@Oracle:localhost> alias CURRENCY SELECT * FROM TAFJV_FBK_CURRENCY
```

To execute this SQL alias switch to SQL mode first.



```
tafj@Oracle:localhost> SQL
```

Then you can execute alias.

```
tafj@Oracle:localhost> CURRENCY
```

You have also the ability to complete an alias with an additional statement, i.e. you want to add criteria to a SQL statement.

```
GVAL0912028.europe.temenosgroup.com - PuTTY
```

RECID	THE_RECORD	CURRENCY_CODE	RANK	NUMERIC_CCY_CODE	CCY_NAME	NO_OF_DE
GBP		GBP	1	130	POUND STERLINGS	12
GOL		GOL	15	6	BULLION1	12
HKD		HKD		150	HONK KONG DOLLARS	12
INR		INR		310	INDIAN RUPEE	12
JPY		JPY	4	180	JAPANESE YEN	10
KHR		KHR		33	riel (KHR)	12
KRW		KRW		389	KOREAN WON	10
LBP		LBP		270	Lebanese Pounds	12
NZD		NZD		220	New Zealand	12
PLN		PLN		280	Polish Zloty	12
SEK		SEK		240	Swedish Krone	12
XEU		XEU		954	Euro	12
SGD		SGD		250	SINGAPORE DOLLAR	12
USD		USD	10	260	US Dollar1	12
ZAR		ZAR		290	South African Rand	12
AUD		AUD		30	AUSTRALIAN DOLLAR	12
CAD		CAD		50	CANADIAN DOLLAR	12
CHF		CHF	6	60	SWISS FRANCS	12
CNY		CNY		388	Chinese Yuan	12

```
tafj@Oracle:localhost> CURRENCY WHERE RECID='USD'
```

The resulting output will be.



```
GVAL0912028.europe.temenosgroup.com - PuTTY
DBTools  SQL  P1:1 C1:12
-----
RECID|THE_RECORD|CURRENCY_CODE|RANK|NUMERIC_CCY_CODE|CCY_NAME  |NO_OF_DECIMALS|Q
USD  |           |USD           |0   |260           |US Dollar1|2           |
-----
tafj@Oracle:localhost>
```

Remove an alias

To delete an existing alias just type command `unalias <<ALIAS NAME>>`

```
tafj@Oracle:localhost> unalias CURRENCY
```

Please note that aliases are saved at the end of your session so you have the ability to load it the next time you will use DBTools Console. Data are stored in a properties file "DBTools" under TAFJ_HOME/conf.

As mentioned before DBTools Console doesn't rely on a user context, in consequence if many users use it at the same time the last one who quit his session will override the properties file with his settings.



Setup

By typing command setup you will access to some properties allowing you to customize your session.

As mentioned before these properties are stored in the DBTools properties file and are shared between all users.

```
tafj@Oracle:localhost> setup
```

	DBTools	SETUP	P1:1 C1:1
1	MAX ROW RETRIEVED (SQL)	200	
2	MAX RECORDS PER PAGE	20	
3	MAX LINE LENGTH	80	
4	HISTORY SIZE MAX	10	
5	SHOW SESSION LONG NAME	false	
6	SHOW PAGE NUMBER	true	
7	SQL AUTO-COMMIT	false	
8	LOG FILE	>DBToolsLog	

Max row retrieved (SQL)

This property allows you to customize the maximal number of rows that would be retrieved from a SQL SELECT statement.

It prohibits memory exception in case of large result set.

To be considered, value should be an Integer, default value is 200.

Max records per page

This property allows you to customize the maximal number of results that would be displayed in the main panel.



To be considered, value should be an Integer, default value is 20. Make sure your console is correctly sized to display the requested number of records.

Max line length

This property allows you to customize the maximal line length that would be displayed in the main panel.

To be considered, value should be an Integer, default value is 80. Make sure your console is correctly sized to display the full line length.

History size max

This property allows you to customize the maximal number of commands that would be stored in the history.

To be considered, value should be an Integer, default value is 10.

Show session long name

This property allows you to customize the session name that would be displayed in the footer panel. When set to 'true' session long name is applied.

To be considered, value should be a boolean, default value is false.

Show page number

This property allows you to display or not information about page number and column number in header panel.

To be considered, value should be a boolean, default value is true.

SQL auto- commit

This property allows you to enable / disable auto-commit on your database connection. When running SQL statements with auto-commit set to false, your change won't be applied unless you execute a COMMIT statement before exiting.

To be considered, value should be a boolean, default value is false.

Log file

This property allows logging command result to the related log file by providing a log file name. Default value is null, which means command result won't be logged.



Navigation and display

Change page/ column

For results that can't be displayed on a single page and/or a single column (by default 20 lines are being displayed per page and a line is 80 characters long), you have the ability to navigate between pages by using the following commands:

- **f** to go one page forward.
- **b** to go one page backward
- **sr** to scroll right to next column
- **sl** to scroll left to previous column

Change console display

When displaying a result you have the ability to fix the first line and / or the first column of the data being displayed for all the pages and columns.

It's especially useful in case of SQL SELECT where you might want to remain fixed the first line that displays column names, and the first column that displays the record id.

By default DBTools console will split result into columns in the case that the record to be displayed is too large to fit on one unique column. You could disable that functionality to have your record displayed on a single column.

Type the following commands:

- **hc** to enable / disable first fixed column
- **hl** to enable / disable first fixed line
- **cm** to enable / disable column mode



User Management Application

Introduction

tUserMgmt is a User application designed to add/Del/Reset users for use of DBTools and JED applications.

Syntax

The application could be launched from TAFJ_HOME/bin.

```
tUserMgmt << MODE >> << -u username>> << -p password>>
```

where MODE can be :

```
--Add << -u username>> << -p password>> ; Add a new User.
```

```
--Reset << -u username>> << -p password>> ; Reset an existing User.
```

```
--Del << -u username>> ; Delete an existing User.
```

The username and encrypted password gets stored in the property file. With the created username and password one can access DBTools and JED application. The username and password has to be passed along with -u and -p option respectively. The username and password is mandatory if either of them passed as null then message "Missing user credentials, please provide a user and password: -u username -p password" is displayed.

Example:

```
Administrator: Command Prompt
tUserMgmt << MODE >> << -u username>> << -p password>>
where MODE can be :
    --Add << -u username>> << -p password>> ; Add a new User.
    --Reset << -u username>> << -p password>> ; Reset an existing User.
    --Del << -u username>> ; Delete an existing User.
```

Creating a new User:



Administrator: Command Prompt

```
D:\T24_DEV>tUserMgnt --Add -u test -p Temenos_1
```

Administrator: Command Prompt

```
USER CREATION SUCCESSFUL  
D:\T24_DEV>
```

Reset an existing user:

Administrator: Command Prompt

```
D:\T24_DEV>tUserMgnt --Reset -u test3 -p Temenos_3
```

Administrator: Command Prompt

```
USER RESET SUCCESSFUL  
D:\T24_DEV>
```

Deleting an existing User:

Administrator: Command Prompt

```
D:\T24_DEV>tUserMgnt --Del -u test4
```

Administrator: Command Prompt

```
USER DELETION SUCCESSFUL  
D:\T24_DEV>
```

JED Console application



Syntax

TAFJ-JED will be accessed only with authenticated users. Hence username and password has to be passed as parameters to JED. The users to access JED needs to be created using a tUserMgnt.bat/tUserMgnt.sh.

TAFJ-JED application is designed to edit and modify database records by providing their file name and record key.

The JED Console application is made to replace JED GUI application on platforms with possibly no X Window installed. : UNIX, AIX, z/OS.

JED Console should also be faster than the current SWING based application, and built upon TAFJ-Core framework to be independent of any native libraries.

It could be called from DBTools console and in standalone mode too.

Execution – Interactive mode

JED Console could be launched from TAFJ_HOME/bin.

```
tJed [-u <username> -p <password> -cf <configuration>[.properties]] <file  
name> <record key>
```

The targeted database will be the one defined in the default configuration file or in the configuration specified by the -cf option. The username and password has to be passed along with -u and -p option respectively.

Example

To edit the record SYSTEM from SPF file.

```
tJed -u <username> -p <password> F.SPF  
SYSTEM
```

Optional parameters

Fields per page number

By default JED console is configured to display 20 fields per page.

By providing option line: -l <fields per page number> you will be able to change dynamically this property. Please ensure that your terminal is correctly sized to display the number of line required or the data won't be displayed properly.

```
tJed -u <username> -p <password> -l 50  
F.SPF SYSTEM
```



Column names

By default JED console is configured to display a record with no field name.

If connected to T24 and by providing the column name option: **-cn** you will be able to retrieve the record field names if they are defined in the related T24 application.

```
tJed -u <username> -p <password> -cn
```

Help

Before editing your record, you have the ability to display JED console available functionalities by providing the help option: **-help**

```
tJed -u <username> -p <password> -help
```

LOCK

By default JED console will read the requested record with no lock on it. To make a **readu** you have to provide the **LOCK** parameter with value **Y**.

```
tJed -u <username> -p <password> F.SPF
```

JED Graphical mode

By providing option graphical user interface: **-gui** you will be able to launch JED in graphical mode.

```
tJed -u <username> -p <password> -gui F.SPF
```

Execution - script mode

It is also possible to modify a record by using JED console from command line, without opening it. That's what we call "script mode".

To update the record **you just have to add '-s' to the command and know the field(s) number you want to change** and provide the associated values.

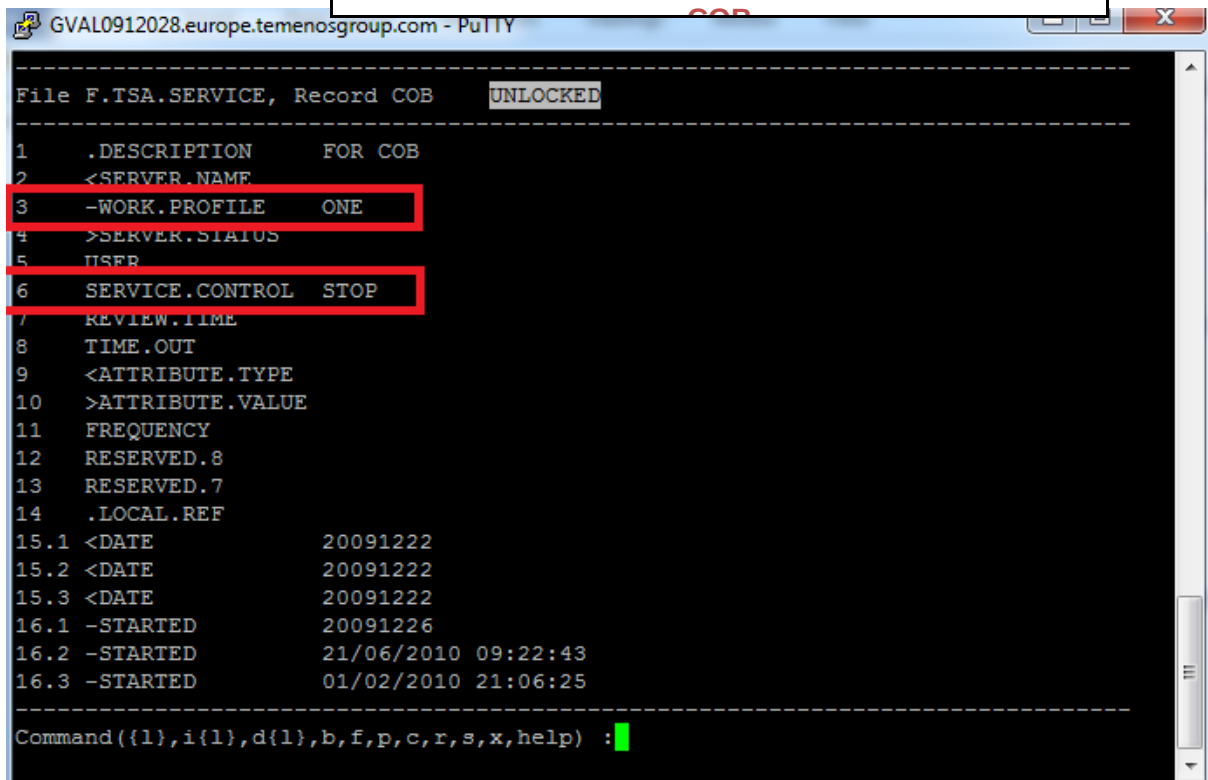
Example

To change in file F.TSA.SERVICE, COB record, field value number of agent (WORK.PROFILE) and service status (SERVICE.CONTROL), you have to do the following:



If you don't have any idea of the related field numbers, firstly you have to open the record in console mode and get these numbers.

tJed -u <username> -p <password> F.TSA.SERVICE



```
File F.TSA.SERVICE, Record COB UNLOCKED
-----
1  .DESCRIPTION      FOR COB
2  <SERVER.NAME
3  -WORK.PROFILE     ONE
4  >SERVER.STATUS
5  USER
6  SERVICE.CONTROL   STOP
7  REVIEW.TIME
8  TIME.OUT
9  <ATTRIBUTE.TYPE
10 >ATTRIBUTE.VALUE
11 FREQUENCY
12 RESERVED.8
13 RESERVED.7
14 .LOCAL.REF
15.1 <DATE           20091222
15.2 <DATE           20091222
15.3 <DATE           20091222
16.1 -STARTED        20091226
16.2 -STARTED        21/06/2010 09:22:43
16.3 -STARTED        01/02/2010 21:06:25
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :
```

We can see that the field numbers we want to update are 3 and 6.

Then you can type the following command to change these fields to “TWO” and “START” :

tJed -u <username> -p <password> -s F.TSA.SERVICE COB
3=TWO;6=START

Note: depending on the operating system you could have to escape the part of command after -s option with quotes: JED -s 'F.TSA.SERVICE COB 3=TWO;6=START;'

The record will be automatically updated to its new value without opening the console.

It also works for multi-value and sub-value. i.e. tJed -u <username> -p <password> -s MY.FILE MYRECORD 1.1=VM;2.1.1=SM;



Presentation

JED console consists of three panels:

- Header panel that contains information about the record
- Main panel that contains the record to be edited
- Footer panel that contains the command line

The screenshot shows a terminal window titled 'guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin'. The interface is divided into three horizontal panels, each outlined in red. The top panel, labeled 'Header', contains the text 'File F.SPF, Record SYSTEM' and 'UNLOCKED'. The middle panel, labeled 'Main panel', displays a list of 20 record fields with their values, such as '1 RUN.DATE 20000729' and '2 SITE.NAME EBS LONDON.'. The bottom panel, labeled 'Footer', shows the command prompt 'Command({1},i{1},d{1},b,f,p,c,r,s,x,help) :'. The 'Main panel' is the largest and contains the majority of the data.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin

File F.SPF, Record SYSTEM  UNLOCKED

1  RUN.DATE      20000729
2  SITE.NAME     EBS LONDON.
3  OP.MODE       0
4  .OP.CONSOLE   TEST
5  <MAIN.ACCOUNT  ../TestBase.data
6  -BACKUP.CYCLE.1
7  >BACKUP.CYCLE.2
8  CURRENT.RELEASE 200912.005
9  HIST.LIFE      5
10 ALL.PG.INC
11 CACHE.EXPIRY
12 ENQ.PAGE.LIMIT 200
13 RECV.DATE.TIME
14 SYS.BACKUP.MODE DISK
15 HOLD.BATCH.OUTPUT Y
16 MICROFICHE.OUTPUT N
17 REPORT.RETENTION 100
18 RESTRICT.RPT.VIEW
19 DATA.ACC.NAME  ../TestBase.data
20 RUN.ACC.NAME    ../TestBase.run

Command({1},i{1},d{1},b,f,p,c,r,s,x,help) :
```

Header

The header presents information related to the edited record:

- File name
- Record key



- Record lock status : the value could be UNLOCKED (default) / LOCKED depending on LOCK parameter provided

```
-----
File F.SPF, Record SYSTEM UNLOCKED
-----
```

File name Record key Record lock status

```
-----
File F.SPF, Record SYSTEM LOCKED
-----
```

- Record status: the value could be **empty** (default, that means no current change made to the record) / **WAITING** (when editing a field, waiting for user entry) / **PENDING** (change made to a record but not validated into database yet) / **SAVED** (change made to the record validated in database).

Record status

```
-----
File F.SPF, Record SYSTEM LOCKED WAITING
-----
1 >20000729<
```

```
-----
File F.SPF, Record SYSTEM LOCKED PENDING
-----
1 20000730
```

```
-----
File F.SPF, Record SYSTEM LOCKED SAVED
-----
```

- Console display mode : value could be empty (default, that means large record will be displayed on multiple line and on a single column) / **COLUMN MODE** (that means large record will be displayed on single line and multiple column)

```
-----
File F.SPF, Record SYSTEM UNLOCKED
-----
1 20100107
2 SINGLE COMPANY TRADE DATED SINGLE COMPANY TRADE DATED SINGLE COMPANY TRADE
2 DATED SINGLE COMPANY TRADE DATED SINGLE COMPANY TRADE DATED SINGLE COMPAN
2 Y TRADE DATED
```

```
-----
File F.SPF, Record SYSTEM UNLOCKED COLUMN MODE
-----
1 20100107
2 SINGLE COMPANY TRADE DATED SINGLE COMPANY TRADE DATED SINGLE COMPANY TRADE
3
```



Main panel

The main panel contains the record to be displayed. The record is displayed vertically and is split into lines. Each line has a number that correspond to a field, a multi-value (i.e. 1.1) or a sub-value (i.e. 1.1.1).

The field number is displayed on a margin on the left part of the panel; the associated value is displayed next to the margin on the middle of the panel.

JED console can display 80 characters per row, by default the margin is 5 characters long and the data part 75 characters long.

When connected to T24 it's possible to display in the margin the field names if they are defined, by providing **-cn** option.

In that case the margin length will increase and the data length will decrease.

```
1 20000729
2 EBS LONDON.
3 O
4 TEST
5 ../TestBase.data
6
7
8 200912.005
9 5
10
11
12 200
13
14 DISK
15 Y
16 N
17 100
18
19 ../TestBase.data
20 ../TestBase.run
```



1	RUN.DATE	20000729
2	SITE.NAME	EBS LONDON.
3	OP.MODE	0
4	.OP.CONSOLE	TEST
5	<MAIN.ACCOUNT	../TestBase.data
6	-BACKUP.CYCLE.1	
7	>BACKUP.CYCLE.2	
8	CURRENT.RELEASE	200912.005
9	HIST.LIFE	5
10	ALL.PG.INC	
11	CACHE.EXPIRY	
12	ENQ.PAGE.LIMIT	200
13	RECV.DATE.TIME	
14	SYS.BACKUP.MODE	DISK
15	HOLD.BATCH.OUTPUT	Y
16	MICROFICHE.OUTPUT	N
17	REPORT.RETENTION	100
18	RESTRICT.RPT.VIEW	
19	DATA.ACC.NAME	../TestBase.data
20	RUN.ACC.NAME	../TestBase.run

Margin

Data

Footer

The footer panel contains the command line that shows some of the available commands. It can also contain some instructions related to console current mode.

Classic mode, waiting for user command

```
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :█
```

Editing mode, waiting for user validation

```
-----
Press ENTER to return to command line
```

Functionalities

Change a record

Edit an existing field / Multi-value / Sub-value

To edit an existing field, multi-value or sub-value just type the field, VM, SM number on the command line, even if the field isn't on the current page.

The prompt will be placed on the related page and line and you will be able to change the field value.



The current field value is placed into bracket that allows you to see if there are blank characters in the current value.

To validate the change, type “enter” and you will return to command line.

You will see the record status changing from WAITING to PENDING during this process.

```

guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM      UNLOCKED
-----
1  RUN.DATE          20000729
2  SITE.NAME         EBS LONDON.
3  OP.MODE           0
4  .OP.CONSOLE       TEST
5  <MAIN.ACCOUNT      ../TestBase.data
6  -BACKUP.CYCLE.1
7  >BACKUP.CYCLE.2
8  CURRENT.RELEASE   200912.005
9  HIST.LIFE         5
10 ALL.PG.INC
11 CACHE.EXPIRY
12 ENQ.PAGE.LIMIT    200
13 RECV.DATE.TIME
14 SYS.BACKUP.MODE   DISK
15 HOLD.BATCH.OUTPUT Y
16 MICROFICHE.OUTPUT N
17 REPORT.RETENTION  100
18 RESTRICT.RPT.VIEW
19 DATA.ACC.NAME    ../TestBase.data
20 RUN.ACC.NAME      ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :23
  
```

```

guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM      UNLOCKED  WAITING
-----
21 DICT.ACC.NAME      ../TestBase.dict
22 AUTO.MANUAL.BATCH  MANUAL
23 OPERATING.SYSTEM   >UNIX<
24 .UNIX.BACKUP
25 .UNIX.RESTORE
26 .PRODUCT.ACCOUNT
27 HELD.RPT.RETENTION 5
28 CACHE.SIZE         500
29 MASTER.ACCOUNT
30 PREVIOUS.RELEASE   200911
31 .PATCHES.INSTLD
32 BATCH.SESIONS
33 <COMP.MNE.CCY
34 >MIN.ROUNDING
35.1 .NON.2DEC.CCY    BEF
35.2 .NON.2DEC.CCY    ESP
35.3 .NON.2DEC.CCY    ITL
35.4 .NON.2DEC.CCY    JPY
35.5 .NON.2DEC.CCY    PTE
35.6 .NON.2DEC.CCY    LUF
-----
Press ENTER to return to command line
  
```



```
guest@chdsk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
21  DICT.ACC.NAME      ../TestBase.dict
22  AUTO.MANUAL.BATCH  MANUAL
23  OPERATING.SYSTEM  AIX
24  .UNIX.BACKUP
25  .UNIX.RESTORE
26  .PRODUCT.ACCOUNT
27  HELD.RPT.RETENTION 5
28  CACHE.SIZE        500
29  MASTER.ACCOUNT
30  PREVIOUS.RELEASE  200911
31  .PATCHES.INSTLD
32  BATCH.SESIONS
33  <COMP.MNE.CCY
34  >MIN.ROUNDING
35.1 .NON.2DEC.CCY      BEF
35.2 .NON.2DEC.CCY      ESP
35.3 .NON.2DEC.CCY      ITL
35.4 .NON.2DEC.CCY      JPY
35.5 .NON.2DEC.CCY      PTE
35.6 .NON.2DEC.CCY      LUF
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) : █
```

Insert a field / Multi-value / Sub-value

To insert a new field, multi-value or sub-value just type the command **i** followed by the field, VM, SM number on the command line.

The prompt will be placed on the related page and line and you will be able to insert the field value.

To validate the insertion, type “enter” and you will return to command line.

You will see the record status changing from WAITING to PENDING during this process.



i.e. **i 2** to insert a field at the second position in the current record.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  SAVED
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12      200
13
14      DISK
15      Y
16      N
17      100
18
19      ../TestBase.data
20      ../TestBase.run
-----
Command({l},i{1},d{1},b,f,p,c,r,s,x,help) :i 2
```

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  WAITING
-----
1      20000729
2      NEW FIELD 2
3      EBS LONDON.
4      O
5      TEST
6      ../TestBase.data
7
8
9      200912.005
10     5
11
12
13     200
14
15     DISK
16     Y
17     N
18     100
19
20     ../TestBase.data
-----
Press ENTER to return to command line
```



i.e. **i 3.3** to insert a third multi-value in the third field of the record. If the third field isn't a multi-value yet, its current value will be set as multi-value 1 and multi-value 2 and 3 will be added.

```

guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2      NEW FIELD 2
3      EBS LONDON.
4      O
5      TEST
6      ../TestBase.data
7
8
9      200912.005
10     5
11
12
13     200
14
15     DISK
16     Y
17     N
18     100
19
20     ../TestBase.data
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :i 3.3
  
```

```

guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  WAITING
-----
1      20000729
2      NEW FIELD 2
3.1    EBS LONDON.
3.2
3.3    ADD VM 3 to FIELD 3
4      O
5      TEST
6      ../TestBase.data
7
8
9      200912.005
10     5
11
12
13     200
14
15     DISK
16     Y
17     N
18     100
-----
Press ENTER to return to command line
  
```

i.e. **i 4.1.2** to insert a second sub-value in the first multi-value of the fourth field of the record. If the fourth field isn't a multi-value neither a sub-value yet, its current value will be set as multi-value 1, sub-value 1 and sub-value 2 will be added.



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM      UNLOCKED  WAITING
-----
1      20000729
2      NEW FIELD 2
3.1    EBS LONDON.
3.2
3.3    ADD VM 3 to FIELD 3
4.1.1  0
4.1.2  ADD SM 2 to FIELD 4.1
5      TEST
6      ../TestBase.data
7
8
9      200912.005
10     5
11
12
13     200
14
15     DISK
16     Y
17     N
-----
Press ENTER to return to command line
```

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM      UNLOCKED  WAITING
-----
1      20000729
2      NEW FIELD 2
3.1    EBS LONDON.
3.2
3.3    ADD VM 3 to FIELD 3
4.1.1  0
4.1.2  ADD SM 2 to FIELD 4.1
5      TEST
6      ../TestBase.data
7
8
9      200912.005
10     5
11
12
13     200
14
15     DISK
16     Y
17     N
-----
Press ENTER to return to command line
```



Delete a field / Multi-value / Sub-value

To delete a field, multi-value or sub-value just type the command **d** followed by the field, VM, SM number on the command line.

The fields, multi-values, sub-values that follow the deleted field will be renumbered.

You will see the record status changing to PENDING during this process.

i.e. **d 2** to delete field number 2

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2      NEW FIELD 2
3.1    EBS LONDON.
3.2
3.3    ADD VM 3 to FIELD 3
4.1.1  O
4.1.2  ADD SM 2 to FIELD 4.1
5      TEST
6      ../TestBase.data
7
8
9      200912.005
10     5
11
12
13     200
14
15     DISK
16     Y
17     N
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :d 2
```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2.1    EBS LONDON.
2.2
2.3    ADD VM 3 to FIELD 3
3.1.1  0
3.1.2  ADD SM 2 to FIELD 4.1
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :|
```

i.e. **d 2.2** to delete multi-value 2 of field 2

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2.1    EBS LONDON.
2.2
2.3    ADD VM 3 to FIELD 3
3.1.1  0
3.1.2  ADD SM 2 to FIELD 4.1
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :d 2.2|
```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2.1    EBS LONDON.
2.2    ADD VM 3 to FIELD 3
3.1.1  O
3.1.2  ADD SM 2 to FIELD 4.1
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :
```

i.e. **d 3.1.2** to delete sub-value 2 of multi-value 1 of field 3

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2.1    EBS LONDON.
2.2    ADD VM 3 to FIELD 3
3.1.1  O
3.1.2  ADD SM 2 to FIELD 4.1
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :d 3.1.2
```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2.1    EBS LONDON.
2.2    ADD VM 3 to FIELD 3
3.1.1  O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :
```

Delete a record

You can delete a record by typing command **d**.

You will be asked for confirmation.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2.1    EBS LONDON.
2.2    ADD VM 3 to FIELD 3
3.1.1  O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :d
```



```

guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1  RUN.DATE          20000729
2.1 SITE.NAME        EBS LONDON.
2.2 SITE.NAME        ADD VM 3 to FIELD 3
3.1.1 OP.MODE        0
4  .OP.CONSOLE       TEST
5  <MAIN.ACCOUNT      ../TestBase.data
6  -BACKUP.CYCLE.1
7  >BACKUP.CYCLE.2
8  CURRENT.RELEASE    200912.005
9  HIST.LIFE          5
10 ALL.PG.INC
11 CACHE.EXPIRY
12 ENQ.PAGE.LIMIT     200
13 RECV.DATE.TIME
14 SYS.BACKUP.MODE    DISK
15 HOLD.BATCH.OUTPUT  Y
16 MICROFICHE.OUTPUT  N
17 REPORT.RETENTION   100
18 RESTRICT.RPT.VIEW
19 DATA.ACC.NAME     ../TestBase.data
-----
Command({l},i{1},d{1},b,f,p,c,r,s,x,help) :d
Are you sure to delete this record Y/N ? █

```

Validation

Cancel pending changes

You have the ability to cancel all your pending modifications by typing command **c**. The record will be restored at its initial state or at its state just after your last save. You will see the record status changing from PENDING to empty during this process.

```

guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1  20000729
2.1 EBS LONDON.
2.2 ADD VM 3 to FIELD 3
3.1.1 0
4  TEST
5  ../TestBase.data
6
7
8  200912.005
9  5
10
11
12  200
13
14  DISK
15  Y
16  N
17  100
18
19  ../TestBase.data
-----
Command({l},i{1},d{1},b,f,p,c,r,s,x,help) :c █

```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
20     ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :█
```

Save pending changes

To update your pending modifications into database, you can use command **s**.

You will see the record status changing from PENDING to SAVED.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  PENDING
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
20     ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :s█
```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  SAVED
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
20     ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) : █
```

Refresh data

The command **r** will allow you to reload the record from the database, that way you will be able to see external modification that have been applied to the record since you have loaded it.

The record status will be set to its default empty value.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  SAVED
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
20     ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :r█
```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
File F.SPF, Record SYSTEM UNLOCKED
-----
1      20000730
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
20     ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) : █
```

Navigation and display

Change page

For record that can't be display on a single page (by default 20 lines are being displayed per page), you have the ability to navigate between pages by using the following commands:

- **f** : will make you move to the next page forward
- **b** : will make you go move previous page backward

i.e. **f** to move one page forward



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED
-----
1      20000730
2      EBS LONDON.
3      0
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
20     ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :f
```

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED
-----
21     ../TestBase.dict
22     MANUAL
23     UNIX
24
25
26
27     5
28     500
29
30     200911
31
32
33
34
35.1   BEF
35.2   ESP
35.3   ITL
35.4   JPY
35.5   PTE
35.6   LUF
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :b
```

i.e. **b** to move to previous page



Change column name mode

If you launched JED console with **-cn** option you will be able to enable and disable the field name display in the margin on left part of the screen.

By default field name display will be enabled.

You can disable it by using the **cn** command.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED
-----
1  RUN.DATE          20000729
2  SITE.NAME         EBS LONDON.
3  OP.MODE           0
4  .OP.CONSOLE       TEST
5  <MAIN.ACCOUNT      ../TestBase.data
6  -BACKUP.CYCLE.1
7  >BACKUP.CYCLE.2
8  CURRENT.RELEASE   200912.005
9  HIST.LIFE         5
10 ALL.PG.INC
11 CACHE.EXPIRY
12 ENQ.PAGE.LIMIT    200
13 RECV.DATE.TIME
14 SYS.BACKUP.MODE   DISK
15 HOLD.BATCH.OUTPUT Y
16 MICROFICHE.OUTPUT N
17 REPORT.RETENTION  100
18 RESTRICT.RPT.VIEW
19 DATA.ACC.NAME     ../TestBase.data
20 RUN.ACC.NAME       ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :cn
```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED
-----
1      20000729
2      EBS LONDON.
3      0
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12     200
13
14     DISK
15     Y
16     N
17     100
18
19     ../TestBase.data
20     ../TestBase.run
-----
Command({l},i{1},d{1},b,f,p,c,r,s,x,help) : █
```



Change column mode

By default JED console will split into as many lines as necessary, fields that are too large to be displayed on a single line.



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
File F.SPF, Record SYSTEM  UNLOCKED  SAVED
-----
1      20000729
2      EBS LONDON.
3      0
4      TEST
5      ../TestBase.data
6      FIELD TOO LARGE TO BE DISPLAYED ON A SINGLE LINE, IT WILL BE SPLIT INTO MA
6      NY LINES BY DEFAULT, BUT IT COULD ALSO BE DISPLAYED ON A SINGLE LINE WHEN
6      USING COLUMN MODE
7
8      200912.005
9      5
10
11
12      200
13
14      DISK
15      Y
16      N
17      100
18
-----
Command({l},i{1},d{1},b,f,p,c,r,s,x,help) :cm
```

You could then pass into COLUMN MODE with command **cm**, the field will be displayed on a single line and multiple columns.

Please be sure to save your pending changes before switching to column mode, as console display will be reinitialized all unsaved modifications will be lost when changing mode.

A flag in top right of the console indicates that column mode is on.

You can now scroll right and left to visualize the entire field, by using commands **sr** and **sl**.



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  SAVED  COLUMN MODE
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6      FIELD TOO LARGE TO BE DISPLAYED ON A SINGLE LINE, IT WILL BE SPLIT INTO MA
7
8      200912.005
9      5
10
11
12      200
13
14      DISK
15      Y
16      N
17      100
18
19      ../TestBase.data
20      ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :sr
```

i.e. use command **sr** to scroll right to next column, and so on until the end of field.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  SAVED  COLUMN MODE
-----
1
2
3
4
5
6      NY LINES BY DEFAULT, BUT IT COULD ALSO BE DISPLAYED ON A SINGLE LINE WHEN
7
8
9
10
11
12
13
14
15
16
17
18
19
20
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :sr
```

You can then use command **sl** to go back to previous column



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  SAVED  COLUMN MODE
-----
1
2
3
4
5
6      USING COLUMN MODE
7
8
9
10
11
12
13
14
15
16
17
18
19
20
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :sl
```

By typing **cm** command you will disable the column mode and get back to the classic mode.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  SAVED  COLUMN MODE
-----
1
2
3
4
5
6      NY LINES BY DEFAULT, BUT IT COULD ALSO BE DISPLAYED ON A SINGLE LINE WHEN
7
8
9
10
11
12
13
14
15
16
17
18
19
20
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :cm
```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED  SAVED
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6      FIELD TOO LARGE TO BE DISPLAYED ON A SINGLE LINE, IT WILL BE SPLIT INTO MA
6      NY LINES BY DEFAULT, BUT IT COULD ALSO BE DISPLAYED ON A SINGLE LINE WHEN
6      USING COLUMN MODE
7
8      200912.005
9      S
10
11
12      200
13
14      DISK
15      Y
16      N
17      100
18
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :█
```

Display help

An integrated help that sum up the available command can be reached by using **help** command.

Type “enter” to exit help.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      S
10
11
12      200
13
14      DISK
15      Y
16      N
17      100
18
19      ../TestBase.data
20      ../TestBase.run
-----
Command({l},i{l},d{l},b,f,p,c,r,s,x,help) :help█
```



```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
Type {line number}      to edit the record at specified line, i.e. 1.1 for VM
Type i {line number}    to insert a record at specified line, i.e. i 1.1.1 for SM
Type d {line number}    to delete the record at the specified line
Type r                  to reload data
Type d                  to delete the whole record
Type f                  to go forward to next page
Type b                  to go backward to previous page
Type p                  to go from page to page
Type cm                 to enable / disable column mode
Type cn                 to display / hide column names if available
Type sl                 to scroll left in column mode
Type sr                 to scroll right in column mode
Type c                  to cancel modifications
Type s                  to save modifications
Type x                  to exit
-----
Type enter to exit help
```

Exit JED Console application

Simply type the exit command **x** to quit. All unsaved modification will be lost.

```
guest@chdisk-tafjdev1: ~/JN/TAFJ_R11GA/bin
-----
File F.SPF, Record SYSTEM  UNLOCKED
-----
1      20000729
2      EBS LONDON.
3      O
4      TEST
5      ../TestBase.data
6
7
8      200912.005
9      5
10
11
12      200
13
14      DISK
15      Y
16      N
17      100
18
19      ../TestBase.data
20      ../TestBase.run
-----
Command({l},i{1},d{1},b,f,p,c,r,s,x,help) :x
```



DBCheck

Introduction

DBCheck is part of the TAFJ-DB tools. It allows checking your database integrity by selecting from the TAFJ_VOC all the tables, and for each, it will do the following:

- Open the table and check its type
- Write a test record on it
- Read that test record
- Read that test record and lock it
- Delete the test record

These operations are being displayed on the console with the format:

Table name - Table type - W(write) R(read) RU (read and lock) D (delete) - Time elapsed

Syntax

The DBCheck syntax is the following:

```
DBCheck [-cf <configuration>[.properties]]
```

The targeted database will be the one defined in your default configuration file or in the configuration specified by the -cf option.

Example

```
DBCheck
```

The resulting output looks like this:



DB Check					
1) FBNK.AA.ARR.ACTIVITY.API\$HIS	Type =XML	WRRUD	407 ms	[DONE]	
2) FEU1.AA.ARR.ACTIVITY.API\$NAU	Type =XML	WRRUD	31 ms	[DONE]	
3) FBNK.AA.ARR.ACTIVITY.CHARGES	Type =XML	WRRUD	31 ms	[DONE]	
4) FMF1.AA.ARR.ACTIVITY.CHARGES\$NAU	Type =XML	WRRUD	47 ms	[DONE]	
5) FSG1.AA.ARR.ACTIVITY.CHARGES\$SIM	Type =XML	WRRUD	47 ms	[DONE]	
6) FEU1.AA.ARR.ACTIVITY.MAPPING\$NAU	Type =XML	WRRUD	47 ms	[DONE]	
.....					
17673) FSG1.SC.FEE.GROUP.HIST	Type =XML	WRRUD	31 ms	[DONE]	
17674) FSG1.SC.FEE.GROUP.HIST\$NAU	Type =XML	WRRUD	16 ms	[DONE]	
17675) FSG1.SC.FEES.MODIFY	Type =XML	WRRUD	15 ms	[DONE]	
17676) FSG1.SC.FUND.FLOW	Type =XML	WRRUD	32 ms	[DONE]	

DBComparer

Introduction

DBComparer is part of the TAFJ-DB tools. It compares the structure and data from two databases defined in the configuration file: the main database and the comparer. It will stop as soon as it will find a difference and it will indicate it in the console.

Requirements

As for the targeted database, the following properties must be set to define the comparer:

- temn.tafj.jdbc.url.comparer
- temn.tafj.jdbc.driver.comparer
- temn.tafj.jdbc.username.comparer
- temn.tafj.jdbc.password.comparer

The following options are available to customize the process:

- ignorefiles : A comma separated list of files without spaces to ignore in the comparison.
- files : A comma separated list of files without spaces to include (without whole VOC) in the comparison
- startwithfile : All of the files before the file specified in the VOC will be skipped.
- threads : Number of threads to start (default is 1).

Syntax

The DBComparer syntax to perform a full comparison is the following:



DBComparer

Example

To run a comparison that starts at the mentioned file in the VOC.

```
DBComparer -startswithfile FBNK.CURRENCY
```

In case of difference in the structure the resulting output looks like this:

```
-----DBComparer-----  
Voc Record Key: %DATA.ACC.VOC  
Voc Record Key: %DATA.ACCOUNT  
Voc Record Key: %DICTIONARY  
Voc Record Key: %OTHER.VOC  
Unable to correctly select VOC record from comparer database: %OTHER.VOC
```

In case of difference in the data the resulting output looks like this:

```
-----DBComparer-----  
pool-1-thread-1 file: (-1) F.TT.TELLER.HP.VERSIONS Checking record: 1  
pool-1-thread-1 file: (-1) F.TSA.WORKLOAD.PROFILE$HIS Checking record: 1  
pool-1-thread-1 file: (-1) F.TSA.WORKLOAD.PROFILE Checking record: 1  
java.lang.RuntimeException: DB::SSELECT::Comparer returned count 214 where  
other returned count 219  
Comparer list:  
Other list: F.TSA.STATUS  
cancelling other threads.
```



TAFJ Promoted Columns

This section explains how to promote columns for TAFJ for performance improvements. Promoted columns are relational columns where data is generated via a function when it is inserted/updated. Therefore, these columns will perform better when indexed over an xml index. **NOTE THAT THIS WILL ONLY WORK FOR SINGLE VALUE COLUMNS. To index multi-value columns see Oracle's XMLINDEX clause. The advanced USE CASE shows how to index a multi-value without XMLINDEX, but it requires a code change that probably wouldn't work with a normal T24 use case. An example of an XMLINDEX for Oracle is:**

```
create index IX_FBNK_CUSTOMER_NAME_1_3 ON FBNK_CUSTOMER(XMLRECORD)
INDEXTYPE IS XDB.XMLINDEX PARAMETERS ('PATHS (INCLUDE (/row/c3))');
```

if you want to index the multi-values in attribute c3 of the xml document.

The below SQL mode command outputs a script to create a promoted column for indexing. Use it with the SPOOL command.

```
CREATE-PROMOTED-COL <<FUNCTION NAME>> ATTRIBUTE=<<Att# in Dict>>
DATATYPE=<<( NUMBER|VARCHAR)>> LENGTH=<<Length for Col>>COLNAME=<<COL
to index>> TABLENAME=<<table>>
```

Here is an example:

```
tafj@localhost:TB201507>SPOOL
```

```
tafj@localhost:TB201507>CREATE-PROMOTED-COL IX_RANK ATTRIBUTE=2
DATATYPE=VARCHAR LENGTH=30 COLNAME=RANK TABLENAME=FBNK_CURRENCY
```

```
tafj@localhost:TB201507>SPOOL
```

The resulting script should show up under \$TAFJ_HOME/log/DBTools.

Below are examples of what to do for each database.

Promoting Columns for SQLServer example

Below is an example of how to promote the RANK column (xml attribute 1) on the FBNK_CURRENCY table

```
set QUOTED_IDENTIFIER on;
go
```

```
CREATE FUNCTION udf_RANK_CURRENCY_C1 (@xmlrecord XML)
RETURNS integer
WITH SCHEMABINDING
BEGIN
```



```
RETURN @xmlrecord.value('(/row/c1/text())[1]', 'integer')
END
```

```
ALTER TABLE FBK_CURRENCY
ADD RANK AS dbo.udf_RANK_CURRENCY_C1(XMLRECORD) PERSISTED

CREATE INDEX ix_FBK_CURRENCY_RANK ON FBK_CURRENCY(RANK)
```

Promoting Columns for DB2 example

```
drop function extractC1_INT@
```

```
create function extractC1_INT(xmlrecord XML)
returns INTEGER
language sql contains sql
no external action deterministic
return xmlcast(xmlquery('$t/row/c1' passing xmlrecord as "t") as varchar(10))@
```

```
set integrity for FBK_CURRENCY off@
```

```
alter table FBK_CURRENCY add RANK INTEGER generated always as
(extractC1_INT(XMLRECORD))@
```

```
set integrity for FBK_CURRENCY immediate checked force generated @
```

```
create index IX_FBK_CURRENCY_RANK on FBK_CURRENCY(RANK)@
```

Promoting Columns for Oracle example

```
alter table fbk_currency
add (
RANK number(10) as (NVL(CAST(extractValue(xmlrecord,'/row/c1') as NUMBER),0))
);
```

```
create index IX_FBK_CURRENCY_RANK on FBK_CURRENCY(RANK)
```

Drop and Recreate the Views (All Databases)

Drop and recreate view so that the new column will go directly to the rdbms column and not the XML column when the view is queried. (View can be retrieved from the particular database or generated by using DBImport with DBImport logging set to DEBUG in \$TAFJ_HOME/conf/TAFJTrace.properties. Here you just want to regenerate the VIEW ONLY for the PARTICULAR TABLE ONLY)

Oracle example of getting the view definition from the database:

```
SQL> set long 100000
SQL> select text from all_views where view_name = 'TAFJV_FBK_CURRENCY';
```

This is a DB2 example only for recreating the view to use the new column RANK.

```
CREATE VIEW TAFJV_FBK_CURRENCY as
SELECT a.RECID, a.XMLRECORD "THE_RECORD"
```



```

.a.RECID "CURRENCY_CODE"
.RANK "RANK"
,XMLCAST(XMLQUERY('$d/row/c2[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"NUMERIC_CCY_CODE"
,XMLCAST(XMLQUERY('$d/row/c3[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "CCY_NAME"
,XMLQUERY('$d/row/c3' passing a.XMLRECORD as "d") "CCY_NAME_3"
,XMLCAST(XMLQUERY('$d/row/c4[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "NO_OF_DECIMALS"
,XMLCAST(XMLQUERY('$d/row/c5[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "QUOTATION_CODE"
,XMLCAST(XMLQUERY('$d/row/c6[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "QUOTATION_PIPS"
,XMLCAST(XMLQUERY('$d/row/c7[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "DAYS_DELIVERY"
,XMLCAST(XMLQUERY('$d/row/c8[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "DAYS_FORWARD"
,XMLCAST(XMLQUERY('$d/row/c9[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"INTEREST_DAY_BASIS"
,XMLCAST(XMLQUERY('$d/row/c10[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"RATE_ALLOWANCE"
,XMLCAST(XMLQUERY('$d/row/c11[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "FIXING_DATE"
,XMLCAST(XMLQUERY('$d/row/c12[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"CURRENCY_MARKET"
,XMLQUERY('$d/row/c12' passing a.XMLRECORD as "d") "CURRENCY_MARKET_12"
,XMLCAST(XMLQUERY('$d/row/c13[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"QUOTATION_SUSP"
,XMLQUERY('$d/row/c13' passing a.XMLRECORD as "d") "QUOTATION_SUSP_13"
,XMLCAST(XMLQUERY('$d/row/c14[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "MID_REVAL_RATE"
,XMLQUERY('$d/row/c14' passing a.XMLRECORD as "d") "MID_REVAL_RATE_14"
,XMLCAST(XMLQUERY('$d/row/c15[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"DEFAULT_SPREAD"
,XMLQUERY('$d/row/c15' passing a.XMLRECORD as "d") "DEFAULT_SPREAD_15"
,XMLCAST(XMLQUERY('$d/row/c16[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "BUY_RATE"
,XMLQUERY('$d/row/c16' passing a.XMLRECORD as "d") "BUY_RATE_16"
,XMLCAST(XMLQUERY('$d/row/c17[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "SELL_RATE"
,XMLQUERY('$d/row/c17' passing a.XMLRECORD as "d") "SELL_RATE_17"
,XMLCAST(XMLQUERY('$d/row/c18[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"UPTO_SMALL_AMT"
,XMLQUERY('$d/row/c18' passing a.XMLRECORD as "d") "UPTO_SMALL_AMT_18"
,XMLCAST(XMLQUERY('$d/row/c19[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"TRSY_SMALL_SPRD"
,XMLQUERY('$d/row/c19' passing a.XMLRECORD as "d") "TRSY_SMALL_SPRD_19"
,XMLCAST(XMLQUERY('$d/row/c20[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"CUST_SMALL_SPRD"
,XMLQUERY('$d/row/c20' passing a.XMLRECORD as "d") "CUST_SMALL_SPRD_20"
,XMLCAST(XMLQUERY('$d/row/c21[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "NEGOTIABLE_AMT"
,XMLQUERY('$d/row/c21' passing a.XMLRECORD as "d") "NEGOTIABLE_AMT_21"
,XMLCAST(XMLQUERY('$d/row/c22[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"TRSY_MED_SPREAD"
,XMLQUERY('$d/row/c22' passing a.XMLRECORD as "d") "TRSY_MED_SPREAD_22"
,XMLCAST(XMLQUERY('$d/row/c23[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"CUST_MED_SPREAD"
,XMLQUERY('$d/row/c23' passing a.XMLRECORD as "d") "CUST_MED_SPREAD_23"
,XMLCAST(XMLQUERY('$d/row/c24[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "REVAL_RATE"
,XMLQUERY('$d/row/c24' passing a.XMLRECORD as "d") "REVAL_RATE_24"
,XMLCAST(XMLQUERY('$d/row/c25[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "TRSY_LIMIT_AMT"
,XMLQUERY('$d/row/c25' passing a.XMLRECORD as "d") "TRSY_LIMIT_AMT_25"
,XMLCAST(XMLQUERY('$d/row/c26[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"MIN_ROUND_AMOUNT"
,XMLCAST(XMLQUERY('$d/row/c27[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"CASH_ONLY_ROUNDING"
,XMLCAST(XMLQUERY('$d/row/c28[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"MIN_ROUND_TYPE"
,XMLCAST(XMLQUERY('$d/row/c29[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"CASH_ROUND_TYPE"
,XMLCAST(XMLQUERY('$d/row/c31[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "PRECIOUS_METAL"
,XMLCAST(XMLQUERY('$d/row/c32[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"EQUIVALENT_CCYS"
,XMLQUERY('$d/row/c32' passing a.XMLRECORD as "d") "EQUIVALENT_CCYS_32"
,XMLCAST(XMLQUERY('$d/row/c33[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "COUNTRY_CODE"
,XMLCAST(XMLQUERY('$d/row/c34[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "FIXED_RATE"
,XMLCAST(XMLQUERY('$d/row/c35[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "FIXED_CCY"
,XMLCAST(XMLQUERY('$d/row/c36[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000))
"FIXED_START_DATE"
,XMLCAST(XMLQUERY('$d/row/c37[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "BASE_CCY_RANK"
,XMLCAST(XMLQUERY('$d/row/c38[position]=1' passing a.XMLRECORD as "d") as VARCHAR(4000)) "AVAILABLE_DATE"

```



```
,XMLCAST(XMLQUERY('$d/row/c39[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "BLOCKED_DATE"
,XMLCAST(XMLQUERY('$d/row/c40[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "CUT_OFF_TIME"
,XMLCAST(XMLQUERY('$d/row/c41[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "CLS_CCY"
,XMLCAST(XMLQUERY('$d/row/c42[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED14"
,XMLCAST(XMLQUERY('$d/row/c43[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED13"
,XMLCAST(XMLQUERY('$d/row/c44[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED12"
,XMLCAST(XMLQUERY('$d/row/c45[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED11"
,XMLCAST(XMLQUERY('$d/row/c46[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED10"
,XMLCAST(XMLQUERY('$d/row/c47[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED9"
,XMLCAST(XMLQUERY('$d/row/c48[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED8"
,XMLCAST(XMLQUERY('$d/row/c49[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED7"
,XMLCAST(XMLQUERY('$d/row/c50[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED6"
,XMLCAST(XMLQUERY('$d/row/c51[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED5"
,XMLCAST(XMLQUERY('$d/row/c52[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED4"
,XMLCAST(XMLQUERY('$d/row/c53[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED3"
,XMLCAST(XMLQUERY('$d/row/c54[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED2"
,XMLCAST(XMLQUERY('$d/row/c55[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RESERVED1"
,XMLCAST(XMLQUERY('$d/row/c56[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "OVERRIDE"
,XMLQUERY('$d/row/c56' passing a.XMLRECORD as "d") "OVERRIDE_56"
,XMLCAST(XMLQUERY('$d/row/c57[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "RECORD_STATUS"
,XMLCAST(XMLQUERY('$d/row/c58[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "CURR_NO"
,XMLCAST(XMLQUERY('$d/row/c59[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "INPUTTER"
,XMLQUERY('$d/row/c59' passing a.XMLRECORD as "d") "INPUTTER_59"
,XMLCAST(XMLQUERY('$d/row/c60[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "DATE_TIME"
,XMLQUERY('$d/row/c60' passing a.XMLRECORD as "d") "DATE_TIME_60"
,XMLCAST(XMLQUERY('$d/row/c61[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "AUTHORISER"
,XMLCAST(XMLQUERY('$d/row/c62[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "CO_CODE"
,XMLCAST(XMLQUERY('$d/row/c63[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "DEPT_CODE"
,XMLCAST(XMLQUERY('$d/row/c64[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "AUDITOR_CODE"
,XMLCAST(XMLQUERY('$d/row/c65[position()=1]' passing a.XMLRECORD as "d") as VARCHAR(4000)) "AUDIT_DATE_TIME"
FROM
"FBNK_CURRENCY" a
```



USE CASE

Here is an advanced example with column SCRIPT.GROUP a multi-value column

Jql

```
SELECT F.SEAT.SCRIPTS WITH SCRIPT.STATUS EQ 'ACTIVE' AND WITH SCRIPT.GROUP EQ 'TB01-START'
```

Sql translation

```
SELECT RECID FROM "TAFJV_F_SEAT_SCRIPTS" WHERE "SCRIPT_STATUS" = 'ACTIVE' and  
( XMLEXISTS('$t/c10[text()='TB01-START']' PASSING "SCRIPT_GROUP_10" as "t") )
```

The above creates a full table scan. Here is one way to index it.

Alter Table for each multi-value

```
ALTER TABLE F_SEAT_SCRIPTS ADD( SCRIPT_GROUP_1 VARCHAR2(4000) as  
( substr(extractValue(xmlrecord,'/row/c10[1]'),1,100) ));  
ALTER TABLE F_SEAT_SCRIPTS ADD( SCRIPT_GROUP_2 VARCHAR2(4000) as  
( substr(extractValue(xmlrecord,'/row/c10[2]'),1,100) ));  
ALTER TABLE F_SEAT_SCRIPTS ADD( SCRIPT_GROUP_3 VARCHAR2(4000) as  
( substr(extractValue(xmlrecord,'/row/c10[3]'),1,100) ));  
ALTER TABLE F_SEAT_SCRIPTS ADD( SCRIPT_GROUP_4 VARCHAR2(4000) as  
( substr(extractValue(xmlrecord,'/row/c10[4]'),1,100) ));  
ALTER TABLE F_SEAT_SCRIPTS ADD( SCRIPT_GROUP_5 VARCHAR2(4000) as  
( substr(extractValue(xmlrecord,'/row/c10[5]'),1,100) ));
```

Create Indexes for each new column

```
CREATE INDEX ix_f_seat_scripts_script_grp_1 ON F_SEAT_SCRIPTS(SCRIPT_GROUP_1);  
CREATE INDEX ix_f_seat_scripts_script_grp_2 ON F_SEAT_SCRIPTS(SCRIPT_GROUP_2);  
CREATE INDEX ix_f_seat_scripts_script_grp_3 ON F_SEAT_SCRIPTS(SCRIPT_GROUP_3);  
CREATE INDEX ix_f_seat_scripts_script_grp_4 ON F_SEAT_SCRIPTS(SCRIPT_GROUP_4);  
CREATE INDEX ix_f_seat_scripts_script_grp_5 ON F_SEAT_SCRIPTS(SCRIPT_GROUP_5);  
CREATE INDEX ix_f_seat_scripts_script_grp_n ON F_SEAT_SCRIPTS (SCRIPT_GROUP_N);
```

Recreate the view with new columns

```
CREATE OR REPLACE VIEW TAFJV_F_SEAT_SCRIPTS as SELECT a.RECID, a.XMLRECORD  
"THE_RECORD"  
,a.RECID "SCRIPT_ID"  
,extractValue(a.XMLRECORD,'/row/c1[position()=1]') "DESCRIPT"  
,extract(a.XMLRECORD,'/row/c1') "DESCRIPT_1"  
,extractValue(a.XMLRECORD,'/row/c2[position()=1]') "COMPANY_CODE"  
,extractValue(a.XMLRECORD,'/row/c3[position()=1]') "SCRIPT_STATUS"  
,extractValue(a.XMLRECORD,'/row/c4[position()=1]') "SCRIPT_SOURCE"  
,extractValue(a.XMLRECORD,'/row/c5[position()=1]') "ALTERNATE_REF"  
,extractValue(a.XMLRECORD,'/row/c6[position()=1]') "SELECT_ROUTINE"  
,extractValue(a.XMLRECORD,'/row/c7[position()=1]') "BASE_RELEASE"  
,extractValue(a.XMLRECORD,'/row/c8[position()=1]') "PRODUCT_GROUP"  
,extractValue(a.XMLRECORD,'/row/c9[position()=1]') "PRODUCT_CODE"  
,extractValue(a.XMLRECORD,'/row/c10[position()=1]') "SCRIPT_GROUP"  
,SCRIPT_GROUP_N "SCRIPT_GROUP_10"  
,SCRIPT_GROUP_1  
,SCRIPT_GROUP_2  
,SCRIPT_GROUP_3  
,SCRIPT_GROUP_4  
,SCRIPT_GROUP_5
```



```
,extractValue(a.XMLRECORD,'/row/c11[position()=1]') "APPLICATION"
,extractValue(a.XMLRECORD,'/row/c12[position()=1]') "STATIC_SETUP"
,extractValue(a.XMLRECORD,'/row/c13[position()=1]') "VERSION"
,extractValue(a.XMLRECORD,'/row/c14[position()=1]') "FUNCTION"
,extractValue(a.XMLRECORD,'/row/c15[position()=1]') "TXN_ID"
,extractValue(a.XMLRECORD,'/row/c16[position()=1]') "FIELD_NAME"
,extract(a.XMLRECORD,'/row/c16') "FIELD_NAME_16"
,extractValue(a.XMLRECORD,'/row/c17[position()=1]') "FIELD_VALUE"
,extract(a.XMLRECORD,'/row/c17') "FIELD_VALUE_17"
,extractValue(a.XMLRECORD,'/row/c18[position()=1]') "FIELD_INPUT"
,extract(a.XMLRECORD,'/row/c18') "FIELD_INPUT_18"
,extractValue(a.XMLRECORD,'/row/c19[position()=1]') "SEAT_ID"
,extractValue(a.XMLRECORD,'/row/c20[position()=1]') "UPDATE_APPL"
,extract(a.XMLRECORD,'/row/c20') "UPDATE_APPL_20"
,extractValue(a.XMLRECORD,'/row/c21[position()=1]') "UPDATE_SCRPT_ID"
,extract(a.XMLRECORD,'/row/c21') "UPDATE_SCRPT_ID_21"
,extractValue(a.XMLRECORD,'/row/c22[position()=1]') "UPD_APPL_FLD_IDEN"
,extract(a.XMLRECORD,'/row/c22') "UPD_APPL_FLD_IDEN_22"
,extractValue(a.XMLRECORD,'/row/c23[position()=1]') "GENERATE_ERROR"
,extractValue(a.XMLRECORD,'/row/c24[position()=1]') "DEFINE_ERROR"
,extractValue(a.XMLRECORD,'/row/c25[position()=1]') "CREATED_BY"
,extractValue(a.XMLRECORD,'/row/c26[position()=1]') "APPROVED_BY"
,extractValue(a.XMLRECORD,'/row/c27[position()=1]') "USER"
,extractValue(a.XMLRECORD,'/row/c28[position()=1]') "MSG_FROM_FILE"
,extractValue(a.XMLRECORD,'/row/c29[position()=1]') "TPR_FILE_NAME"
,extractValue(a.XMLRECORD,'/row/c30[position()=1]') "CREATED_DATE"
,extractValue(a.XMLRECORD,'/row/c31[position()=1]') "LAST_MODIFIED_DATE"
,extractValue(a.XMLRECORD,'/row/c32[position()=1]') "IO_ANALYSIS"
,extract(a.XMLRECORD,'/row/c32') "IO_ANALYSIS_32"
,extractValue(a.XMLRECORD,'/row/c33[position()=1]') "RESERVED_2"
,extractValue(a.XMLRECORD,'/row/c34[position()=1]') "RESERVED_1"
,extractValue(a.XMLRECORD,'/row/c36[position()=1]') "OVERRIDE"
,extract(a.XMLRECORD,'/row/c36') "OVERRIDE_36"
,extractValue(a.XMLRECORD,'/row/c37[position()=1]') "RECORD_STATUS"
,extractValue(a.XMLRECORD,'/row/c38[position()=1]') "CURR_NO"
,extractValue(a.XMLRECORD,'/row/c39[position()=1]') "INPUTTER"
,extract(a.XMLRECORD,'/row/c39') "INPUTTER_39"
,extractValue(a.XMLRECORD,'/row/c40[position()=1]') "DATE_TIME"
,extract(a.XMLRECORD,'/row/c40') "DATE_TIME_40"
,extractValue(a.XMLRECORD,'/row/c41[position()=1]') "AUTHORISER"
,extractValue(a.XMLRECORD,'/row/c42[position()=1]') "CO_CODE"
,extractValue(a.XMLRECORD,'/row/c43[position()=1]') "DEPT_CODE"
,extractValue(a.XMLRECORD,'/row/c44[position()=1]') "AUDITOR_CODE"
,extractValue(a.XMLRECORD,'/row/c45[position()=1]') "AUDIT_DATE_TIME"
FROM
"F_SEAT_SCRIPTS" a
```

Change query

```
SELECT RECID FROM TAFJV_F_SEAT_SCRIPTS WHERE "SCRIPT_STATUS" =
'ACTIVE' and (SCRIPT_GROUP_1 = 'TB01-START' OR SCRIPT_GROUP_2 = 'TB01-
START' OR SCRIPT_GROUP_3 = 'TB01-START' OR SCRIPT_GROUP_4 = 'TB01-START'
OR SCRIPT_GROUP_5 = 'TB01-START');
```